BANK OF THAILAND

Monetary Policy Report

July 2013

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
July 2013

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Monetary Policy in Thailand

The Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprise of the Governor and two deputy Governors, as well as four distinguished external members representing various sectors of the country, with the aim to provide monetary policy decisions that are effective and transparent.

The Monetary Policy Objective

The MPC sets monetary policy with the objective to support sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

The Monetary Policy Target

The Cabinet approves the use of quarterly average core inflation with the range of 0.5-3.0 percent as its policy target for 2013, as proposed by the MPC together with the Minister of Finance. In the event that the target is missed, the MPC is required to explain the reasons and resolution thereof to the public.

The Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the key policy rate to signal the monetary policy stance.

Evaluating economic conditions and forecasts

The Bank of Thailand considers information from all sources, the macroeconomic model, data in each economic sector, as well as direct opinions from large enterprises and small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and covers all aspects, both in the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of Monetary Policy communication to the public, the MPC arrange many forms of communication, both in Thai and English, such as (1) organizing a press statement at 14.00 hrs. on the day of the meeting, (2) publishing Minutes of the MPC meeting two weeks after the meeting, and (3) publishing the Monetary Policy Report every quarter.

Monetary Policy Report

July 2013

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Growth and Inflation Prospects and Monetary Policy

1. Growth and Inflation Prospects and Monetary Policy

The Thai economy was expected to grow at a slower pace than previously assessed. Private consumption was likely to moderate after having accelerated sharply in the previous periods driven by government stimulus measures. Export recovery could delay on the back of China's economic slowdown and domestic supply constraints. Weaker domestic and external demand in turn caused further delays in private investment. Meanwhile, inflationary pressure was expected to ease in line with softening private demand, lower-than-anticipated world prices of oil and non-fuel commodities, as well as additional effects from domestic energy price policies.

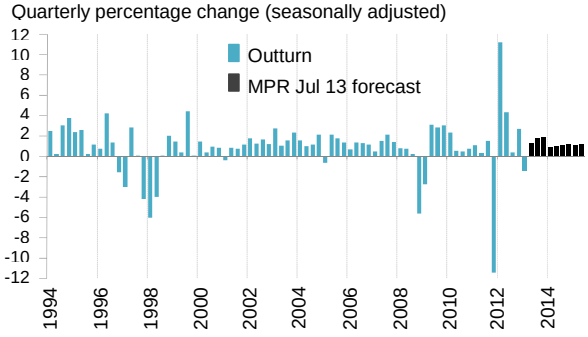
In the past two meetings, the MPC voted to reduce the policy rate by 0.25 percent in the first meeting and hold the policy rate in the latter meeting. The MPC deemed that an accommodative monetary policy stance to help support domestic demand remained appropriate given rapidly changing global economic and financial conditions as well as financial stability risks which warranted close monitoring going forward.

1.1 Growth and inflation prospects

The Thai economy was likely to moderate more than previously projected, while inflationary pressure was poised to ease.

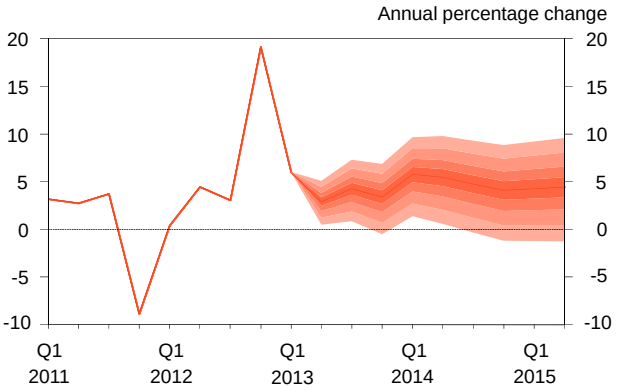
In 2013, the Thai economy was expected to grow at a slower pace than previously projected. Main economic activities were likely to slow down. Private consumption softened from waning effects of the first-car subsidy scheme which came sooner than expected, coupled with rising household debt burden which constrained consumer spending. Meanwhile, private investment moderated in line with lower demand, both domestic and external, while merchandise exports were further

Chart 1.1 Thailand's GDP growth



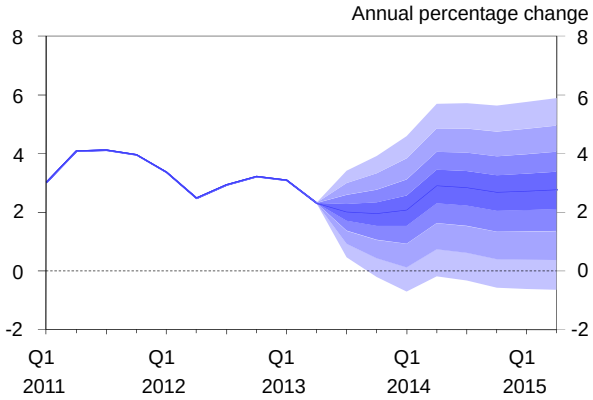
Source: Office of the National Economic and Social Development Board and calculation by Bank of Thailand
Note: At 1988 prices (seasonally adjusted)

Chart 1.2 GDP growth forecast



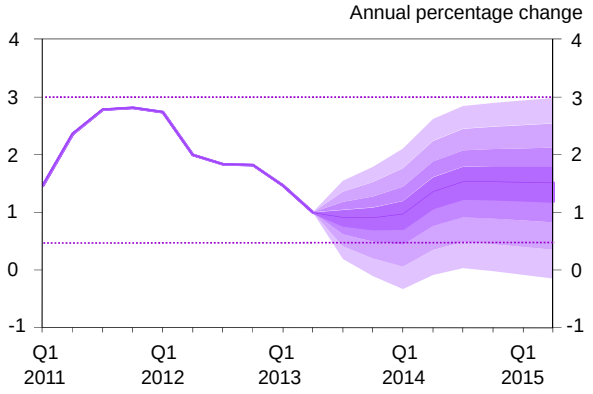
Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.3 Headline inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

Chart 1.4 Core inflation forecast



Note: The fan chart covers 90 percent of the probability distribution.

affected by China's economic slowdown and domestic supply-side problems following the disease outbreak in shrimps. Nevertheless, the MPC assessed that the Thai economy would gain momentum gradually going forward, supported by favorable income and employment conditions, accommodative monetary conditions, both in credit and interest rates, as well as the gradual recovery of the G3 economies, which would help support Thai economic growth in 2014 **(Chart 1.1** and **Table 1.1)**.

The fragile global economy remained the key risk factor that could cause the Thai economy to expand lower than the baseline projection. The MPC viewed that Thailand's trading partners' economies could grow at a slower pace than expected if China's economic slowdown persisted longer than anticipated and caused widespread effects on Asian economies. Nevertheless, positive contributions from continued economic recovery in the US on the back of improving economic fundamentals, and the pick-up in the Japanese economy led by government stimulus measures, would help support overall global economic recovery. However, in the short run, the financial markets could be somewhat volatile due to the anticipation that the Federal Reserve (Fed) could wind down its quantitative easing (QE), which in effect could cause further capital outflows from emerging markets. **Domestic risk factors also shifted to the downside**. There was a possibility that private consumption and investment could slow down by more than expected during the end of 2013 when positive contribution from government stimulus measures would end with the termination of these measures. Furthermore, spending on the water management projects could be delayed on account of the Administrative Court's ruling that a public hearing must be sought on the projects. Overall, the MPC assessed that risks to economic growth would tilt

towards the downside, compared to the previous assessment of balanced risks. The fan chart for growth was thus skewed down throughout the projection period **(Chart 1.2)**.

At the same time, inflation was poised to remain stable at the currently low level with inflationary pressure easing further compared to the previous projection. This stemmed from lower demand pressure on the back of softer private demand, and lower cost pressure following the downward trend of world oil and non-fuel commodity prices. In addition, inflationary pressure also alleviated from the Committee on Energy Policy Administration's (CEPA) decision to postpone the LPG price hike by another two months. Overall, inflation projections were thus lower than previously projected **(Table 1.1)**. Going forward, the MPC assessed that risks to inflation would slightly tilt towards the downside given the possibility that demand pressure could be lower than assessed in the baseline case as downside risks on economic growth outweighing upside risks. Fan charts for inflation were thus skewed downwards slightly, compared to those that were previously skewed up **(Charts 1.3 and 1.4)**.

Table 1.1 Forecast summary

Percent	2012*	2013	2014
GDP growth	6.5	4.2	5.0
		(5.1)	(5.0)
Headline inflation	3.0	2.3	2.6
		(2.7)	(2.7)
Core inflation	2.1	1.1	1.4
		(1.6)	(1.7)

Note: * Outturn
( ) Monetary Policy Report April 2013
Source : Office of the National Economic and Social Development Board and calculation by Bank of Thailand

1.2 Economic outlook

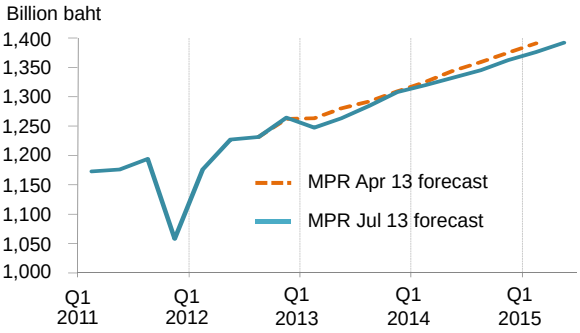
The Thai economy was likely to grow less than previously projected due to moderate private spending, possible delays in export recovery and lower government stimulus than previously expected.

The Thai economy was poised to moderate

Thai economic growth was expected to slow down given that the main engine of growth, namely private demand, declined substantially from the previous projection. Economic indicators in April and May suggested that economic activities continued to slow down from the first quarter of the year. On this front, positive effects from the first-car tax rebate scheme on private consumption faded sooner than expected. Private investment moderated significantly on the back of weaker demand, both domestic and external. This was consistent with manufacturing production which was expected to experience year-on-year contraction. Meanwhile, fiscal stimulus was likely to be lower than previously assessed in tandem with lower-than-expected budget disbursements of local governments coupled with disbursement delays in the water management projects. **With regard to the external sector, exports were still poised to recover gradually**, but possibly at a slower pace than previously assessed owing to the more-than-expected slowdown of Chinese and Asian economies. Nonetheless, imports of goods and services were also expected to decelerate in line with softer exports and private investment. As a result, net exports as a whole would contribute positively to economic growth more than previously assessed.

Weakening domestic and external demand momentum could lead to lower Thai economic growth (Chart 1.5 and Table 1.2), especially in 2013 where economic growth projections were adjusted downward quite significantly **(Table 1.1)**. However, going forward, momentum of the overall economy was expected to improve gradually from the following supporting factors: (1) high employment and income, (2) investment demand by businesses as part of their production restructuring plans and in accordance with the government's large-scale investment projects, and (3) expected gradual recovery of export income in tandem with the recovering trend of the G3 economies. These positive factors would help boost Thai economic growth in 2014 to grow close to the rate previously projected.

Chart 1.5 Level of GDP



Source: Office of the National Economic and Social Development Board and calculation by Bank of Thailand
Note: At 1988 prices (seasonally adjusted by the Bank of Thailand)

Table 1.2 Forecasts for GDP and components

Percent	2013	2014
GDP growth	4.2	5.0
Domestic demand	3.9	5.1
Private consumption	3.4	3.5
Private investment	2.0	9.9
Government consumption	4.2	3.5
Public investment	15.6	8.3
Exports of goods and services	5.8	8.0
Imports of goods and services	5.8	8.4

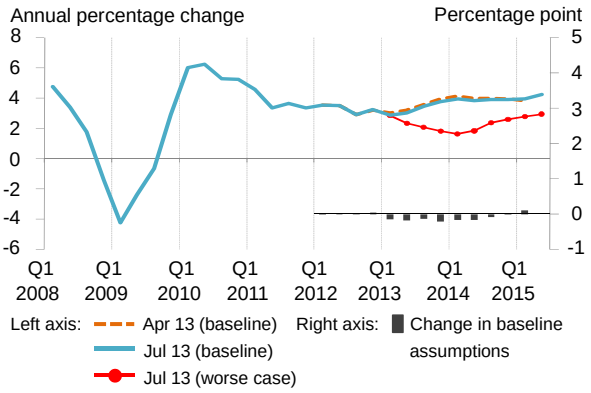
Note: At 1988 prices

The global economy should recover gradually

Overall, the global economy would continue to recover slowly, but at a slower pace than previously assessed due mainly to China and Asia's economic slowdown.

Thailand's trading partners' economies were expected to recover gradually, although the momentum of recovery would be slightly lower

Chart 1.6 Growth assumptions for Thailand's trading partners



Note: Weighted by each country's share in Thailand's total exports

Chart 1.7 Growth assumptions for the US

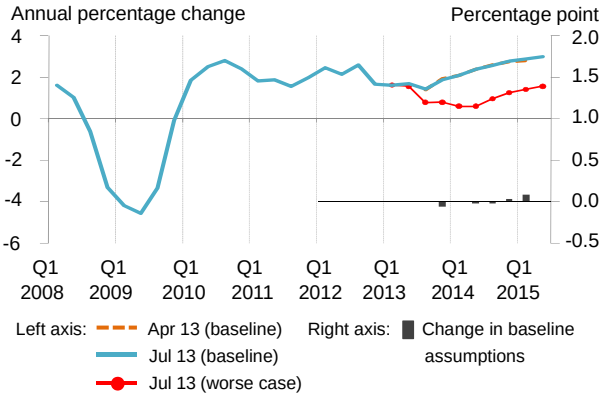
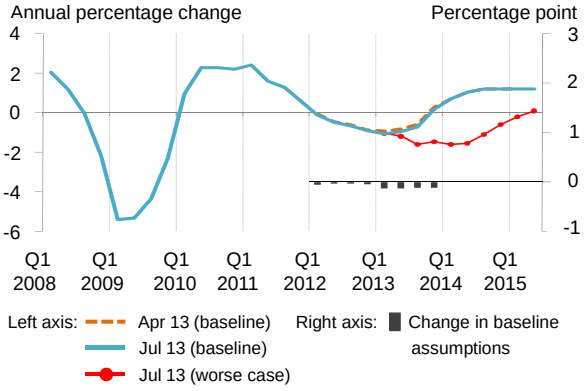


Chart 1.8 Growth assumptions for the euro area



than that previously projected (Chart 1.6). The main drag was the greater-than-expected slowdown of the Chinese economy, which in turn could cause Asian economies to moderate further. Nevertheless, positive growth momentum from the US economy, which was expected to recover continuously on the back of improving economic fundamentals, and the recovery of the Japanese economy led by government stimulus measures, should support the overall recovery path of Thailand's trading partners' economies to remain intact.

The US economy was likely to improve slowly as previously assessed (Chart 1.7), consistent with improvements in economic fundamentals, especially in the housing sector which showed clear signs of recovery as reflected by rising confidence and house prices. Moreover, the labor market recovered continuously with the unemployment rate trending downwards and non-farm payroll remaining strong. Improvements in US' real economic data were in accordance with the Fed's signal to reduce asset purchases this year which caused short-term volatility in the global financial markets and could lead to an increase in long-term interest rates. Nevertheless, the MPC assessed that a rise in long-term interest rates would not have a large impact on the real economy. Although private spending in the US was likely to recover continuously going forward, reductions in government spending and uncertainties regarding deliberations on the debt ceiling increase would continue to put constraint on US economic recovery in future periods.

Euro area economies became more stabilized and were expected to recover gradually (Chart 1.8). The MPC viewed that euro area economies would recover more concretely during the latter half of 2013 as previously expected, mainly supported by major member economies. The recovery

would be driven mainly by improvements in core member economies such as Germany which started to show signs of improvements in consumption and exports as well as in the manufacturing sector. Nonetheless, fiscal austerity measures implemented in many countries would continue to put constraint on recovery going forward and could affect economic activities more than assessed. The MPC evaluated that the Fed's inclination to gradually reduce asset purchases would not have a significant effect on euro area economies. In addition, the MPC expected that improving US economic conditions and the stronger US dollar trend would somewhat help boost the euro area's exports.

The Japanese economy was likely to expand more favorably in 2013 (Chart 1.9) thanks to domestic spending which benefited from government stimulus measures, both the supplementary budget and continuous monetary policy easing which aimed to help pull Japan out of deflation. The MPC assessed that impacts from these measures would pass through to the real economy more concretely. Private consumption was expected to expand continuously, while the manufacturing sector started to exhibit positive signs as reflected by a pickup in the Manufacturing Production Index and business confidence. However, a more prolonged economic slowdown in Japan's key trading partners-namely China and other regional countries-than expected, could pose risks to the Japanese economy going forward.

Notwithstanding the improvement in Japan's economic outlook, the outlook for Chinese and Asian economies weakened overall (Chart 1.10). China's economy was expected to slow down by more than anticipated by the market due mainly to sluggish exports and moderate investments in the manufacturing sector. In addition, liquidity

Chart 1.9 Growth assumptions for Japan

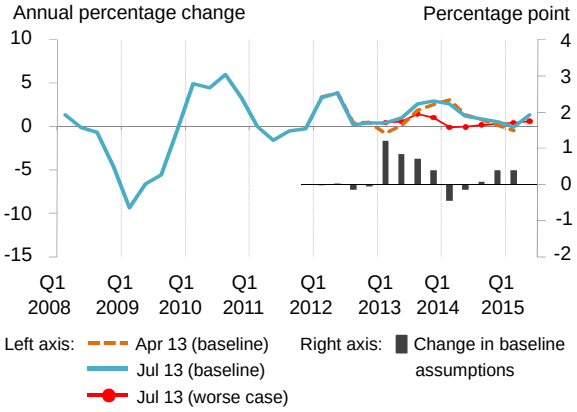
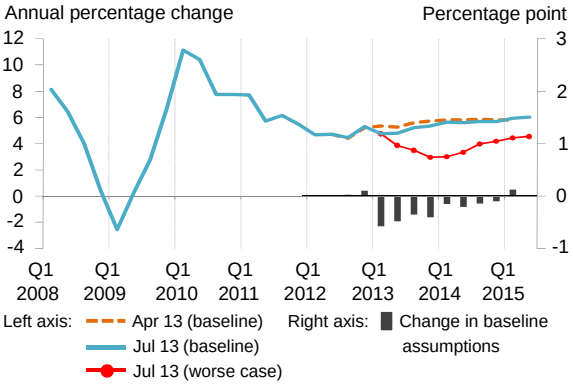


Chart 1.10 Growth assumptions for Asian economies (excluding Japan)



Note: Weighted by each country's share in Thailand's total exports

problems in the financial markets turned more severe, partly resulting from credit control measures and demand for liquidity which was sharply risen. Nevertheless, the manufacturing sector and business investments were expected to improve gradually during the latter half of the year spurred by economic recovery of the G3 countries coupled with strong business profits and government measures to expedite investments. Meanwhile, China's economic slowdown was expected to adversely affect other Asian economies via lower trade and commodity prices. The MPC evaluated that support from economic recovery in the US and Japan would not be enough to fully offset China's economic slowdown. Furthermore, going forward, the MPC viewed that downside risks to Asia's economic growth would increase from the risk that China's economic slowdown could protract for longer than anticipated and from the risk of capital outflows from emerging markets due to the tapering of asset purchases by the Fed.

Thai export recovery could be delayed

Recovery of the Thai export sector was likely to be delayed. Economic recovery in the US was unable to fully offset the economic slowdown in China and Asia. Meanwhile, imports also decelerated in line with exports and investment.

The export sector remained stable during the first half of the year, but a recovery in the next periods was likely to be slower than previously assessed due to the slight deceleration in trading partners' economic recovery (Table 1.3). The MPC assessed that manufacturing exports across most industries, except for automobiles, were poised to slow down going forward. Exports of hard disk drives could moderate in line with softer world demand, while exports of integrated circuits could recover gradually with the economic recovery of the G3

countries. Meanwhile, exports of fishery and agricultural products were expected to be affected by supply shortages due to the disease outbreak in shrimps. Moreover, the expansion in rice exports was likely to be limited on account of deteriorating price competitiveness.

Table 1.3 Forecasts for the external sector

	2013	2014
Growth in value of exports* (F.O.B., percent)	4.0	8.0
Growth in value of imports* (F.O.B., percent)	5.5	8.5
Trade balance (billion U.S. dollars)*	5.4	4.8
Current account balance (billion U.S. dollars)*	1.7	0.1

Note: *Data revision according to definitions in IMF's Balance of Payments Manual, 6th edition (BPM6), and Ministry of Commerce's revised database

The MPC evaluated that positive effects from the US economic recovery was unable to fully offset negative effects from China and Asia's economic slowdown. Thai exports to the US had not yet fully benefited from the US' economic recovery. This could be because improvements in US economic fundamentals would take time to pass through to the real economy. In addition, the disease outbreak in shrimps led to tepid exports of fishery products to the US since the start of the year. Meanwhile, merchandise exports to China could be further affected from the Chinese slowdown, especially those for final consumption in China. However, exports of intermediate goods were expected to expand continuously thanks to the gradual economic recovery of the G3 countries.

At the same time, imports of goods and services were likely to moderate more than previously assessed (Table 1.3). This was consistent with the expected slowdown in private spending and exports which had not yet fully recovered. The import value of machinery and equipment, especially that of export-oriented sectors,

continued to decline. Nevertheless, capital imports in the medium- to long-term should gain support from a gradual pickup in investments of manufacturers to restructure their production process and from the government's large-scale investment projects.

Looking ahead, the current account balance[1/] was likely to record a lower surplus, but remained similar to the previous projection (Table 1.3). This was attributable to improvements in the services account balance following the decline in freight payments due to lower merchandise imports, which helped compensate for the decline in the trade balance from softer export value.

Private demand was poised to soften

Private demand plateaued after accelerating sharply last year. Looking ahead, private demand was expected to moderate from the previous projection.

Private consumption was likely to remain stable after accelerating sharply in the previous periods on the back of government stimulus measures. The MPC assessed that private consumption would expand at a lower rate than previously anticipated due mainly to (1) sooner-than-expected waning of positive impacts from the first-car tax rebate scheme, while higher debt burden from this scheme led households to become more cautious in their spending behavior which would affect consumption spending, especially in durable and semi-durable goods, going forward, and (2) a decline in purchasing power of farm households compared to the previous assessment following lower farm income, partly due to rubber prices which could be adversely affected by the Chinese economic slowdown, while rice prices would continue to benefit

[1/] Including reinvested earnings

from the government's rice pledging scheme for some time. Nevertheless, the MPC viewed that sound economic fundamentals would continue to sustain consumption, and would help support consumption growth to return to normal levels by 2014. These economic fundamentals included (1) strong purchasing power of non-farm households, consistent with tight labor market conditions and high non-farm income, (2) accommodative monetary conditions given low interest rates and continued private credit expansion despite some signs of financial institutions' credit tightening, (3) the depreciation of the baht which would benefit export income in baht terms, and (4) persistently low and stable inflation.

Private investment was expected to expand at a lower rate, particularly in 2013, in line with the export and domestic consumption trends. This partly stemmed from excess capacity in the business sector which led to investment deferrals during periods of weak foreign demand and sluggish domestic demand. Overall, however, investment demand would still remain in order to restructure the production process to replace manual labor replacement. The aforementioned slowdown was consistent with the declining import value of capital goods in the manufacturing sector, particularly export-oriented, which had remained low since early 2013. Nevertheless, the MPC assessed that investment would gain momentum more concretely in 2014, thanks to medium and long-term supporting factors, most importantly (1) the need to restructure the production process to accommodate labor shortage problems, (2) high levels of application and issuance of investment promotion certificates by the Board of Investment (BOI), of which most were found to undertake their investments in accordance to their investment promotion applications, (3) supportive monetary conditions for corporate loan expansion, and

(4) large-scale public investment projects which were expected to crowd in private investment, especially long-term public investment in infrastructure, despite short-term investment delays in the water management projects.

Looking ahead, the MPC viewed that the Thai economy would still be faced with many important structural problems, such as problems regarding the logistic system, water management, as well as the **labor supply problems which are currently the main structural constraints on corporate investment, both in terms of labor shortages and low labor productivity growth**. The MPC assessed that the Thai labor market would remain tight, while labor force growth would be limited. This in turn could affect businesses' investment decisions to expand capacity utilization, as well as put pressure on some entrepreneurs to move their production base to neighboring countries with lower wages. Moreover, Thai labor productivity in previous periods had decelerated sharply from the past, and grew substantially slower than other countries in the region. This was in part a result of labor movement from high productivity economic sectors to low productivity sectors. The MPC viewed that these limitations were long-standing structural problems which would take time to resolve and emphasized that macroeconomic policies that aimed to reduce constraints on the manufacturing sector and to support continued private demand momentum were of essence (more details in BOX: Labor Supply Constraints and Economic Growth).

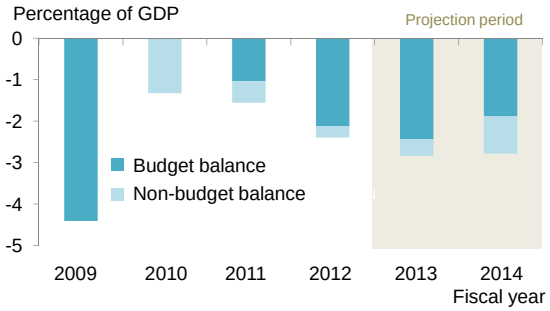
Continuous fiscal stimulus

The public sector would continue to provide economic stimulus. However, overall spending could decline slightly from the previous assessment given risks of delayed disbursements of non-budgetary expenditures.

Fiscal policy would continue to provide economic stimulus going forward as reflected by budget deficit which was relatively high in proportion to GDP. Although the budget deficit was expected to decline from 300 in fiscal year 2013 to 250 in fiscal year 2014, non-budgetary expenditures were likely to increase in the next fiscal year **(Chart 1.11)**.

Compared to the previous projection, fiscal stimulus was likely to decline slightly as a result of deferred disbursements of local governments and postponement of some spending on the water management projects (Table 1.4). Budgetary expenditures declined from both current and capital expenditures, due partly to lower-than-expected disbursements of local governments owing to the delay in disbursements of transfers and allocation of funds from the central government. For non-budgetary expenditures, the MPC reduced its assumption on the amount of funding for the water management projects throughout the projection period to comply with the ruling of the Administrative Court that a public hearing must be sought before proceeding with the projects. This could cause postponement of projects that had not yet been approved. Meanwhile, the assumption on the two trillion baht infrastructure project remained unchanged as the bill drafting process was in accordance with the expected plan.

Chart 1.11 Assumption on public sector expenditure



Note: Non-budgetary spending includes (1) the Mega Investment Project (Strong Thailand), (2) the water management project, (3) the long-term infrastructure investment project, and (4) the Development Policy Loan (DPL)

Source: Public Debt Management Office and Bureau of the Budget and calculation by Bank of Thailand

Table 1.4 Assumptions on public sector expenditure

Unit: Billion baht	Fiscal year	
	2013	2014
General government consumption	1,615.6	1,759.3
Public investment	744.1	855.2
Total	2,359.7	2,614.5

Note: Includes expenditure assumptions on the water management project and the infrastructure investment project

Source: Bureau of the Budget and calculation by Bank of Thailand

Chart 1.12 Output Gap

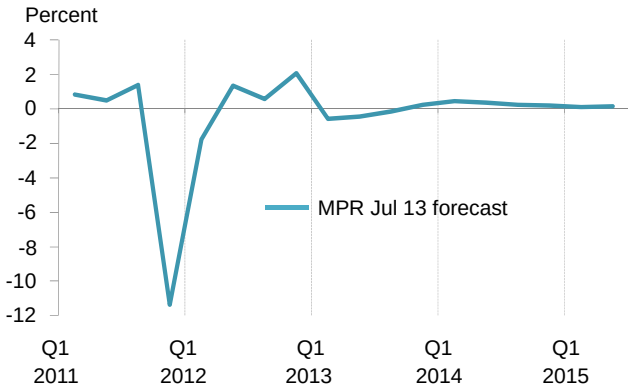
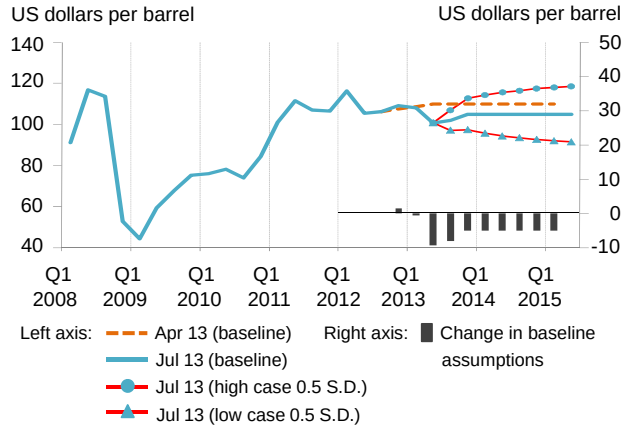


Chart 1.13 Assumptions on Dubai oil price



Inflationary pressure was likely to ease

Inflationary pressure eased continuously since the beginning of the year due to softening demand and lower production costs. Risks to inflation were slightly skewed downwards.

Going forward, inflationary pressure was likely to remain low, with demand pressure easing from sluggish private spending. This was consistent with the output gap remaining close to zero throughout the projection period **(Chart 1.12)**. The MPC assessed that household consumption could be affected by the higher debt burden, while the purchasing power of farm households could decline in line with the subdued farm income trend. Meanwhile, the corporate sector was likely to postpone some investments on the back of delays in export recovery and softening domestic consumption. Lower demand pressure would also lessen the pass through of production costs to prices.

Cost pressure was also likely to soften in line with the expected decline in world oil and non-fuel commodity prices. The MPC slightly reduced the **Dubai oil price** assumption in the baseline case to average at 104 US dollars per barrel for 2013 before stabilizing at 105 US dollars per barrel for the rest of the projection period **(Chart 1.13)**. This adjustment was on account of the somewhat lower global demand trend, mainly from China and Asia. Meanwhile, the supply of world crude oil was likely to increase slightly following higher crude oil production in non-OPEC countries. Overall however, the MPC viewed that world oil prices would remain close to the current level going forward, with demand and supply fundamentals balancing out. As a result, high case and low case assumptions on the Dubai oil price was balanced at half a standard deviation above and below the baseline path. At the same time, pressure from **world**

non-fuel commodity prices also declined **(Chart 1.14)** in tandem with metal prices which fell in line with lower demand from China. The MPC viewed that commodity prices would grow at a low level throughout the projection period, consistent with the gradual global economic recovery. Meanwhile, **fresh food prices** increased somewhat in the short-term from higher-than-expected actual prices and prolonged supply problems **(Chart 1.15)**.

Moreover, Thailand's policy on energy prices would also have an impact on cost pressure. The MPC maintained the assumption of a diesel excise tax exemption throughout the projection period and postponed the hike in LPG prices for another two months for both the household and transport sectors to adhere to the Committee on Energy Policy Administration's (CEPA) decision. This was expected to further ease cost pressure in the short-term, despite higher oil import costs from the relatively weak baht compared to the previous periods.

Looking ahead, the MPC assessed that risks to inflation would tilt slightly towards the downside given the possibility that domestic spending could grow less than expected, which in effect, would lead to lower production cost pass-through.

Chart 1.14 Assumptions on non-fuel commodity prices

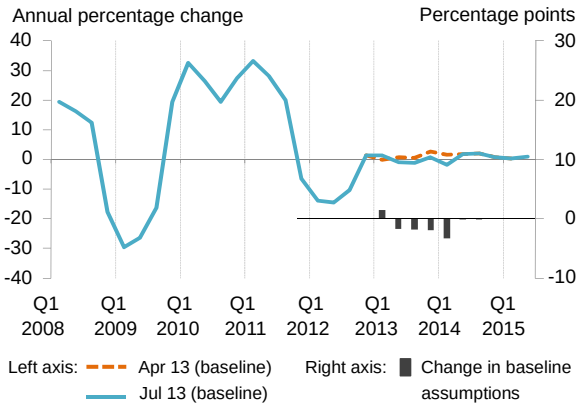
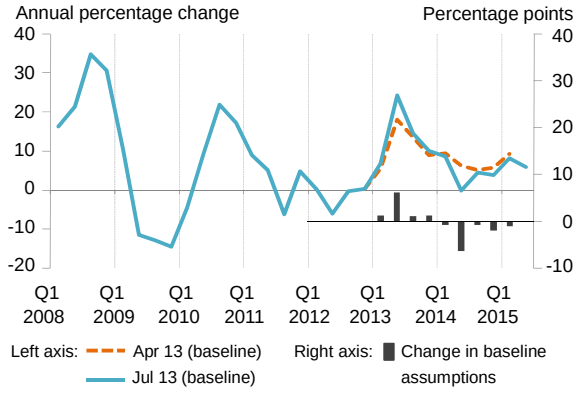


Chart 1.15 Assumptions on fresh food prices



Alternative scenarios

The MPC appraised that there were various scenarios where Thai economic growth could deviate from the baseline case. The MPC thus took these scenarios into consideration to reflect key risk factors on the Thai economy and inflationary pressure going forward.

Scenario 1: The Chinese economic slowdown intensified and protracted more than assessed. There was a possibility that China's exports could continue to grow at a subdued rate should G3

economies recover slower than expected. This in turn could adversely affect China's manufacturing production and cause Asian economic growth to slow down more than anticipated. Furthermore, there was a chance that tight liquidity problems in the financial markets in China could spread out and turn into a systemic risk. Under this scenario, the MPC appraised that Thailand's economic growth could be lower than assessed in the baseline case due to an additional factor of sluggish exports of consumer goods for Chinese consumption. Moreover, world commodity prices could decline further from softer Chinese demand, in particular rubber prices which make up a significant proportion of Thai farmers' income.

Scenario 2: Private consumption grew less than the baseline case. Although the adjustment in private consumption during the first half of the year was expected, the slowdown was more pronounced than anticipated. Moreover, there was a possibility that household spending could be lower than assessed in the baseline case should the higher household debt burden from the first-car scheme cause commercial banks to tighten their lending standards and consumers to exercise more caution in their spending behavior, in particular those in low and medium-income households. This could further affect consumption of durable and semi-durable goods more than expected in the baseline case. In this scenario, the MPC assessed that the Thai economy would expand less than the baseline case from lower spending and postponement of businesses' investment decisions.

Under both scenarios, the MPC appraised that an accommodative monetary policy stance remained appropriate. The MPC would take a wait and see approach and could postpone interest rate normalization for some time.

1.3 Monetary policy decision

Monetary policy stance remained accommodative

In the past two meetings, the MPC voted to reduce the policy rate by 0.25 percent in the first meeting and hold the policy rate in the latter meeting. The MPC deemed that an accommodative monetary policy stance to help support domestic demand remained appropriate given rapidly changing global economic and financial conditions, as well as financial stability concerns which warranted close monitoring.

In its meetings on May 28-29, 2013, the MPC assessed that the global economic recovery was slower than expected due mainly to China and Asia's lower-than-expected economic growth. Meanwhile, global financial markets remained volatile, which could lead to capital flow and exchange rate volatility in the region. In 2013 Q1, The Thai economy grew less than expected from tepid domestic demand, which could weigh on economic momentum going forward. Moreover, exports were subjected to more downside risks from growth moderation in regional economies, especially China. Meanwhile, inflationary pressure eased further owing to lower production costs, while growth of private credit and household debts remained elevated. The MPC judged that the Thai economy should continue to expand on the back of sound economic fundamentals, although downside risks had increased from lower-than-expected growth in the first quarter of this year. As inflation remained well within the target, monetary policy had room to further cushion against downside risks from domestic demand. Against the backdrop of continued financial stability concerns, the MPC thus voted unanimously to reduce the policy rate by 0.25 percent from 2.75 to 2.50 percent per annum. The MPC would closely monitor developments of the Thai economy, financial

stability risks as well as the capital flow situation, and stand ready to take appropriate action as warranted.

In its subsequent meetings on July 9-10, 2013, The MPC appraised that the overall global economy expanded less than assessed in the previous meeting due to China's economic slowdown which weighed on the recovery of Asian exports, despite some improvements in the US economy and stronger Japanese economic growth from government stimulus policies. Growth of the Thai economy moderated from a slowdown in domestic demand and exports. This was partly because consumers exercised more caution in their spending behavior as a result of the rising debt burden, coupled with waning government stimulus measures. Exports decelerated from domestic supply constraints as well as China's slowdown. Weak domestic and external demand in turn caused further delay in private investment. The MPC evaluated that the slowdown in domestic demand was partly an adjustment from the previous extraordinary growth induced by government stimulus measures, and thus should resume its normal growth path in the periods ahead. This was supported by sound underlying fundamentals, as well as accommodative monetary and fiscal policies as reflected in continued private credit growth and fiscal deficits, which would help boost domestic demand growth. Meanwhile, rapidly changing global economic and financial conditions, as well as risks to domestic financial stability warranted close monitoring. The MPC thus voted unanimously to maintain the policy rate at 2.50 percent per annum.

1.4 Appendix:

Table 1.5 Forecast assumptions	2012	2013	2014
Dubai oil price (U.S. dollars per barrel)	109.3	104.0	105.0
Non-fuel commodity prices (%YoY)	-9.8	0.0	0.6
Fresh food prices (%YoY)	-1.5	13.7	4.1
Minimum wage in the Bangkok Metropolitan Region (baht per day)	279	300	300
Government consumption (%YoY)[1/]	10.5	7.6	5.9
Public investment (%YoY)[1/]	12.2	18.8	14.2
Fed Funds rate (% at year-end)	0.13	0.13	0.13
Trading partners' economic growth (%YoY)[2/]	3.3	3.3	3.9
Regional currencies vis-à-vis the US dollar (Index)[3/]	108.4	108.6	107.8

Note: [1/] Including spending on water management plans and infrastructure investment projects
[2/] Weighted by each country's share in Thailand's total exports
[3/] Appreciation against the US dollar indicated by a decrease

Table 1.6 GDP growth forecasts by research houses	2013	2014
HSBC	5.0	6.2
Credit Suisse	4.7	5.5
Nomura	4.7	5.0
Citi	4.7	4.8
UBS	4.7	4.5
NESDB[1/]	4.2-5.2	n.a.
DBS	4.6	5.2
Kiatnakin Bank	4.5	5.0
Tisco Securities	4.5	5.0
FPO[2/]	4.0-5.0	n.a.
BOT	4.2	5.0
Standard Chartered	4.0	5.5
Siam Commercial Bank	4.0	4.9
Kasikorn Research	4.0	4.5

Note: Compiled and published by Reuters on July 10, 2013, except:
[1/] Published on May 20, 2013, with the release of GDP data for 2013 Q1
[2/] Published on June 27, 2013
Presented in descending order of 2013's forecast

Table 1.7 Headline inflation forecasts by research houses		
	2013	2014
DBS	2.9	3.7
NESDB[1/]	2.3-3.3	n.a.
Credit Suisse	2.7	3.4
UBS	2.7	3.0
Kiatnakin Bank	2.7	2.9
Nomura	2.6	2.7
Standard Chartered	2.5	3.2
HSBC	2.5	3.0
Tisco Securities	2.5	3.0
Kasikorn Research	2.5	2.8
Siam Commercial Bank	2.5	2.8
FPO[2/]	2.0-3.0	n.a.
Citi	2.4	2.3
BOT	2.3	2.6

Note: Compiled and published by Reuters on July 10, 2013, except:
[1/] Published on May 20, 2013, with the release of GDP data for 2013 Q1
[2/] Published on June 27, 2013
Presented in descending order of 2013's forecast

Table 1.8 Probability distribution of GDP growth forecast									
Percent	2013			2014				2015	
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 12	0	0	0	0	1	0	0	0	1
10-12	0	0	0	3	3	2	2	2	3
8-10	0	2	1	13	12	9	6	8	9
6-8	1	14	9	27	24	19	16	17	17
4-6	19	36	26	30	27	26	24	24	23
2-4	52	33	34	19	20	23	24	23	22
0-2	26	13	21	7	10	13	16	16	15
(-2)-0	3	2	7	2	3	5	8	7	7
< (-2)	0	0	1	0	1	2	3	3	3

Table 1.9 Probability distribution of headline inflation forecast								
Percent	2013		2014				2015	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 7	0	0	0	1	1	1	1	1
6-7	0	0	0	3	3	2	3	3
5-6	0	1	2	7	7	6	7	7
4-5	1	4	7	14	14	13	13	13
3-4	11	14	16	21	20	19	19	18
2-3	36	28	23	21	21	21	20	20
1-2	37	29	23	17	17	17	17	16
0-1	13	17	16	10	10	12	11	11
(-1)-0	2	6	8	5	5	6	6	6
< (-1)	0	1	4	2	3	3	4	4

Table 1.10 Probability distribution of core inflation forecast								
Percent	2013		2014				2015	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 4.0	0	0	0	0	0	0	0	0
3.5-4.0	0	0	0	0	1	1	1	1
3.0-3.5	0	0	0	1	2	3	3	3
2.5-3.0	0	0	1	5	7	8	8	8
2.0-2.5	0	2	5	12	16	15	15	15
1.5-2.0	6	11	14	21	22	21	21	20
1.0-1.5	33	28	24	24	22	21	21	20
0.5-1.0	43	32	25	19	16	16	16	16
0.0-0.5	16	20	18	11	9	9	9	10
< 0.0	2	8	12	7	5	6	6	7

Recent Economic Developments

2. Recent Economic Developments

The global economy was expected to expand at a more subdued pace in light of China's and Asia's slower growth. This would weigh on Asian export recovery in the period ahead. Meanwhile, G3 economies were expected to improve gradually, with the US economy showing some signs of recovery on the back of improving housing and labor market conditions. The euro area economy started to stabilize, while Japanese economic growth gained better pace thanks to the government's economic stimulus package.

The Thai economy in 2013 Q2 was expected to moderate compared to the same period last year due to softer domestic demand on the back of more cautious household spending and waning fiscal stimulus measures. Exports slowed down as a result of the weakening Chinese economy as well as domestic supply constraints. Weak domestic and external demand in turn caused further delay in private investment. Meanwhile, inflationary pressure eased due to softening domestic demand and lower production costs.

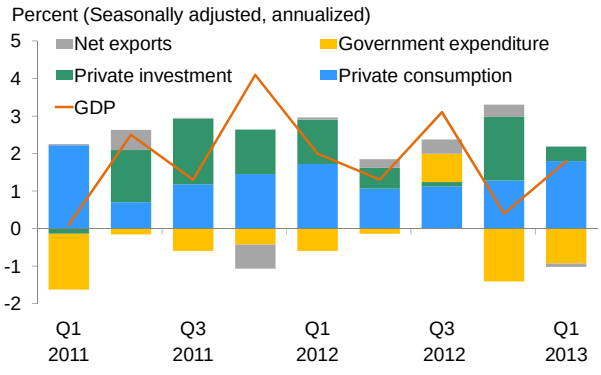
2.1 The global economy

The US economy recovered gradually following improvements in employment and the housing sector which would be important drivers of economic growth going forward.

In 2013 Q1, the US economy expanded by 1.8 percent on a quarter-on-quarter, seasonally adjusted annualized basis (qoq saar) **(Chart 2.1)**, accelerating from the previous quarter which grew 0.4 percent. This was attributable to a pickup in domestic demand, especially from strong private consumption growth following improvements in employment and the housing sector.

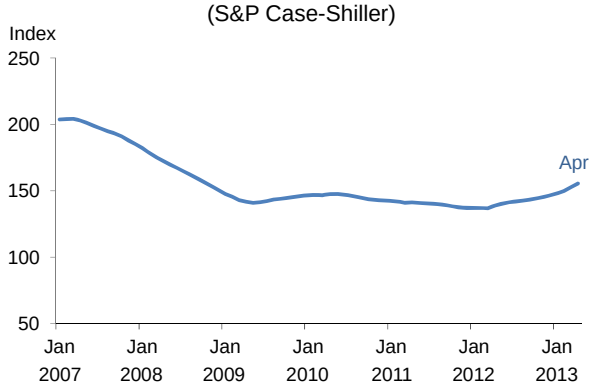
In 2013 Q2, the US economy was expected to slow down from the previous quarter. This was mainly due to budget sequestration which came into effect since March 2013, resulting in public spending

Chart 2.1 Contribution to US economic growth
(Change from the previous quarter)



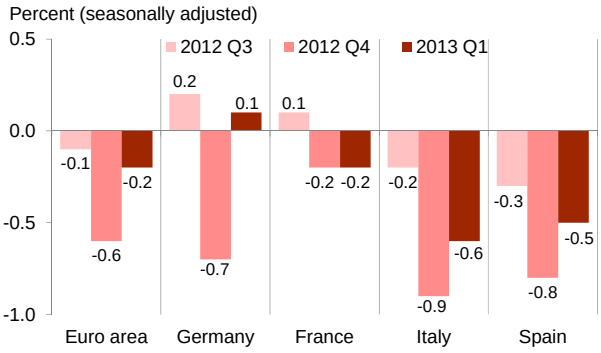
Source: Bureau of Economic Analysis

Chart 2.2 US Housing Price Index
(S&P Case-Shiller)



Source: Bloomberg

Chart 2.3 Euro area's economic growth
(Change from previous quarter)



Source: Eurostat

cuts. Manufacturing and investment in the defense industry would be particularly affected by the budget cuts. At the same time, tax hikes, for example on personal income taxes and payroll taxes, which came into effect in January, would start to weigh on private consumption from this quarter onwards.

Nevertheless, the MPC anticipated that the US economy would continue to show gradual improvements despite the drag of sequestration. This was due to continued improvements in employment conditions and a concrete recovery in the housing sector, in terms of both price and sales **(Chart 2.2)**, which would help improve households' financial position. In addition, continued accommodative monetary policy[1/] in the US would lend support to private consumption growth in the future.

Euro area economies remained weak, but started to show some signs of improvement and was expected to recover gradually in the latter half of 2013.

In 2013 Q1, euro area economies[2/] contracted quarter-on-quarter by 0.2 percent, compared to the previous quarter's contraction of 0.6 percent **(Chart 2.3)**. The German economy showed seasonally

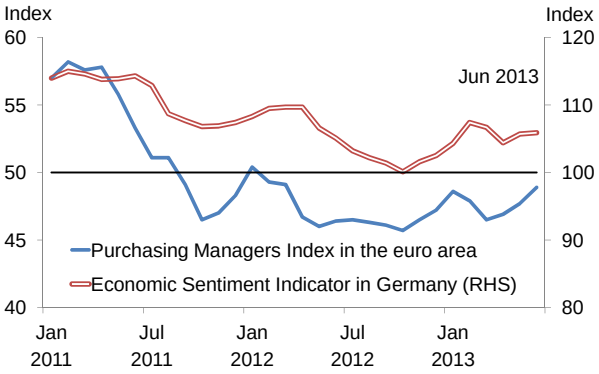
[1/] The Federal Open Market Committee Meeting of the Federal Reserve held in June 2013 announced its decision to reduce the size of asset purchases at end-2013 and could end the asset purchase program in mid-2014 should the economy recover as expected, that is, US economic growth (Central Tendency) of 2.3-2.6 percent and 3.0-3.5 percent in 2013 and 2014, respectively (based on year-on-year growth in the fourth quarter of each year).

[2/] Consists of 17 countries that have adopted the euro as their common currency. The German, French, Italian and Spanish economies accounted for 28, 21, 17 and 11 percent of the euro area economies in 2012, respectively, while Greece, Ireland and Portugal together accounted for 6 percent.

adjusted quarter-on-quarter growth of 0.1 percent in the first quarter, compared to the previous quarter's contraction of 0.7 percent. The improvement was attributable to private consumption growth which had turned positive, while growth in exports and private investment remained negative. Meanwhile, the French economy contracted by 0.2 percent, the same level as the previous quarter, on the back of declining domestic and external demand. The Italian economy contracted by 0.6 percent, while that of the Spanish economy continued to contract by 0.5 percent (qoq sa) owing to fiscal austerity measures and high unemployment.

The second quarter of 2013 saw further economic stabilization, as reflected by continuous growth in manufacturing production and an expansion in export growth to non-euro countries after experiencing contractions in the last 2 months. In particular, exports to the US started to turn positive. At the same time, economic sentiment indicators of the euro area economy and core members improved continuously after the political situation in Italy and the financial crisis in Cyprus subsided. The Purchasing Managers Index adjusted upwards slightly from last quarter's average of 47.6 to stand at 47.8, while Germany's indicator of economic sentiment increased from 105.3 to 106.1 this quarter **(Chart 2.4).**

Chart 2.4 Euro Area Purchasing Managers Index and German Economic Sentiment Indicator



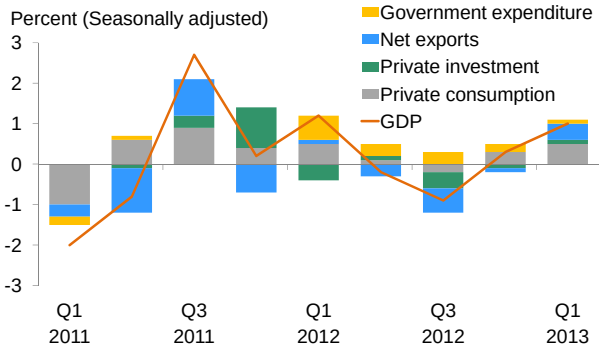
Source: Bloomberg

Looking ahead, the MPC assessed that the euro area economy would gradually recover in the latter half of 2013, driven by the manufacturing sector which started to show signs of improvement and a continuous pickup in private sector confidence. In the longer term, the economy was expected to benefit from greater emphasis placed on economic restructuring with recommendations having been given to member states to focus on enhancing competitive advantage and addressing high

unemployment problems rather than relying only on austerity measures.

The Japanese economy started to show signs of recovery and was poised to expand gradually. Main drivers included private consumption and public spending stemmed mainly from government stimulus measures.

Chart 2.5 Contribution to Japan's economic growth
(Change from the previous quarter)



Source: Cabinet Office of Japan

In 2013 Q1, the Japanese economy grew by 1.0 percent, accelerating from the previous quarter which expanded by 0.3 percent **(Chart 2.5)**. This was mainly attributable to private consumption which benefited from government stimulus measures and a pickup in exports following recovery in demand of trading partners' economies, especially the US.

The Japanese economy in 2013 Q2 was expected to recover continuously thanks to funding from the supplementary budget in 2012 which had begun to enter into the economic system via investments to repair buildings damaged by the 2011 earthquake and maintenance repairs on infrastructure. At the same time, consumer and producer confidence picked up, in part from government stimulus measures and increased business profits from the weaker yen. This in turn, helped boost recovery in private consumption. Meanwhile, manufacturing production was expected to recover gradually, as reflected by the continuous increase in the Manufacturing Production Index since the previous quarter.

Looking ahead, the MPC appraised that the Japanese economy would improve on the back of increased public spending as part of government stimulus package, and private consumption thanks to continuous improvements in consumer confidence. This was reflected in an increase in financial asset prices and the healthier labor market. The continued accommodative monetary policy also helped support

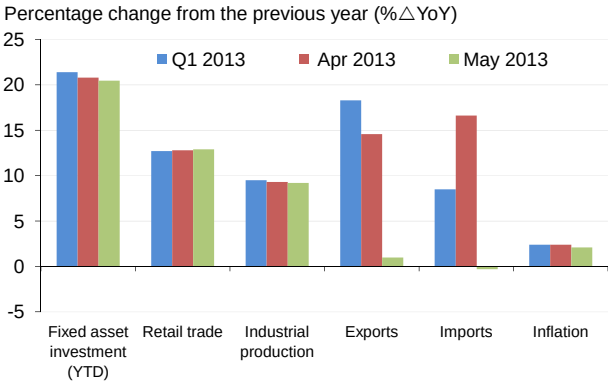
low costs for the business sector. On April 4, 2013, the Bank of Japan announced an expansion of the open-ended asset purchase program in secondary markets and increased the maturity of Japanese government bonds that could be purchased, as well as changed the main operating target for money market operations from an uncollateralized overnight call rate to a monetary base.

Chinese and Asian economies were expected to moderate in 2013 Q2 following the deceleration in the manufacturing and export sectors, while domestic demand continued to play a key role in economic growth.

In 2013 Q1, the Chinese economy grew by 7.7 percent, slowing down from 7.9 percent in the previous quarter, due mainly to a decline in investment. Economic activity moderated in 2013 Q2 compared to the first quarter, following a slowdown in the manufacturing sector, investment and exports, while consumption remained unchanged **(Chart 2.6)**. The substantial decline in exports was in part due to the new rules issued on May 5, 2013 that strengthened the oversight of over-trade invoicing. Going forward, the MPC appraised that the Chinese economy would grow less than anticipated due to lower-than-expected exports and investments, particularly in the first half of 2013. The economy was expected to improve in the second half of 2013 driven by domestic consumption with support from strong employment and income as well as exports which would recover slowly in line with an anticipated pickup in the global economy. Nevertheless, economic growth would not be as high as observed in the past following the Chinese government's policy to give priority to economic stability.

In 2013 Q1, most regional economies grew less than previously expected, which led economic activity in the second quarter to also grow at a slower

Chart 2.6 China's economic indicators



Source: CEIC

rate. North Asian economies continued to moderate in line with softer manufacturing production and investment, while exports remained sluggish despite gradual improvements in some export categories such as electronics, in particular smart phones and communication equipment. In order to support economic recovery, the South Korean and Taiwanese governments announced measures to help stimulate consumption and investment in May 2013. In 2013 Q2, economic activities of ASEAN economies began to show signs of weakening from the previous quarter. This was in part due to persistently sluggish exports, particularly those to China which had moderated continuously since the first quarter. However, strong domestic demand in this region so far had remained an important driver of economic growth.

Looking ahead, the MPC appraised that regional economies would see lower than anticipated growth, due to the more-than-expected economic slowdown in China, one of the region's major trading partner. However, regional economies would gradually improve in the latter half of 2013 in anticipation of continuous domestic demand strength and a slowly recovering export sector in tandem with the global economic recovery, particularly in the US and Japan which would help boost exports to some extent.

2.2 The domestic economy

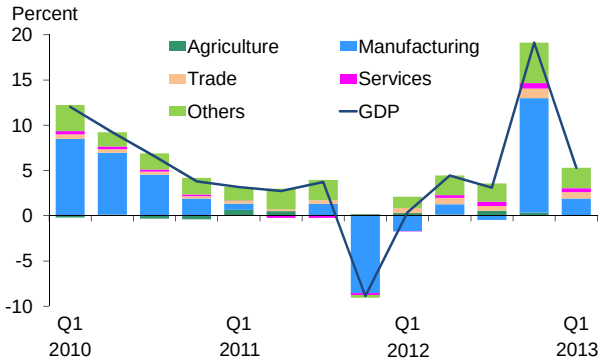
In 2013 Q1, the Thai economy expanded at a slower rate compared to the same period last year, both in terms of expenditures and manufacturing production. Consumption and investment of the private and public sectors slowed down from the previous quarter. At the same time, the gradual

recovery of global economy had yet to offset sluggish domestic demand, which resulted in a moderation in manufacturing overall.

Economic conditions in 2013 Q1

In 2013 Q1, the Thai economy grew year-on year by 5.3 percent **(Chart 2.7)**, slowing down from the previous quarter owing mainly to the deceleration in manufacturing and services sectors. **Manufacturing production** expanded year-on-year by 4.8 percent, slowing from the previous quarter which rose considerably from a low base in 2011 due to the floods. Manufacturing for domestic demand grew at a slower rate compared to the previous quarter in which the automobile production was accelerated to full capacity. Consequently, the production could not increase further until capacity-enhancing investments were to be completed. Meanwhile, manufacturing for exports contracted year-on-year as production of hard disk drives and integrated circuits were affected by sluggish external demand. In addition, production of frozen shrimp declined from a supply shortage following a disease outbreak.

Chart 2.7 Contribution to GDP growth
(compared to the same period last year)



Source: Office of the National Economic and Social Development Board

The services sector expanded year-on-year by 6.7 percent, but slowed down from the previous quarter mainly from the trading sector and the hotel and restaurant sector. This was consistent with the slowdown in domestic demand and the tourism sector as reflected by the number of foreign tourists. At the same time, the construction sector moderated in line with public and private investment and was also affected by higher construction material and labor costs.

Agricultural outputs, overall, grew slightly at 0.5 percent from the same period last year, slowing from the previous quarter due to major agricultural outputs, namely rubber and sugar cane, which

moderated on the back of unfavorable weather conditions. Meanwhile, production of rice, palm oil and cassava continued to expand well. Fishery production contracted as a result of a substantial decline in shrimp output due to the continued outbreak of the early mortality syndrome (EMS).

Continual economic expansion led to a low unemployment rate of 0.7 percent in 2013 Q1, similar to the same period last year. So far, the nationwide minimum wage hike to 300 baht implemented since January 1, 2013, had not shown to cause abnormality in unemployment condition. This was due to continued tightening of the labor market over the past several years. As demand for labor remained high, laid-off workers were thus generally re-employed in other manufacturing sectors. In addition, businesses had already made adjustments to mitigate impacts from higher labor costs.

Demand in 2013 Q1 expanded at a slower rate following last year's sharp acceleration both in consumption and private investment. At the same time, exports slowed down in line with the slow pace of global economic recovery, consistent with the decline in export-oriented manufacturing production.

Table 2.1 GDP growth rate

Change from the previous quarter (seasonally adjusted, percent)	2012					2013
	Year	Q1	Q2	Q3	Q4	Q1
GDP	6.5	0.4	4.4	3.1	19.1	5.3
Domestic demand[1/]	8.4	3.1	6.8	9.0	15.0	4.5
Private consumption	6.7	2.9	5.3	6.0	12.4	4.2
Private investment	14.4	9.2	11.8	16.2	20.9	3.1
Government consumption	7.5	-0.2	7.4	10.0	12.5	2.2
Public investment	8.9	-9.6	4.0	13.2	31.1	18.8
Exports of goods and services	3.1	-3.2	1.1	-2.8	19.6	8.4
Imports of goods and services	6.2	4.3	8.6	-1.8	15.0	8.2

Note: [1/] Domestic demand excluding changes in stocks
Source: National Economic and Social Development Board and calculations by Bank of Thailand

Private consumption expanded year-on-year by 4.2 percent **(Table 2.1)**, decelerating from the previous quarter due to lower demand for non-durable goods, particularly in oil and gas. Demand for durable goods also slowed, in particular that for automobiles, as extraordinary expansion of car orders induced by the first-car tax rebate scheme started to taper off. **Private investment** expanded by 3.1 percent year-on-year, slowing from the previous quarter. Investment in machinery and equipment decelerated considerably after having increased substantially last year to replace and repair flood-related damages. Furthermore, the slow recovery in the global economy led some businesses to postpone their investment plans. Investment in construction also moderated due mainly to a slowdown in residential and office building constructions.

The slow global economic recovery led **exports of goods and services** to expand by merely 8.4 percent, a significant decline compared to the previous quarter, particularly in electronics, metal products, machinery and automobiles. In addition, exports of fishery products also shrunk due to the supply shortage of shrimps. However, exports of agricultural products improved led by rubber and cassava exports. At the same time, exports of services expanded in response to a continuous increase in the number of foreign tourists, particularly from Russia and China. Meanwhile, freight receipts and other transport receipts declined in line with slower international trade.

The slowdown in domestic consumption together with a slow recovery in exports led to a deceleration in **imports of goods and services** which recorded 8.2 percent growth from previous quarter. The moderation was seen across all import categories, in particular, the import of capital goods,

raw materials and intermediate goods such as electronic parts, metal products and petroleum products. Import of services also slowed down from the previous quarter due to a decline in freight payments in line with slower international trade, despite the increase in travel expenditures as more Thais travelled abroad.

The **public sector** expanded at a slower rate both in terms of consumption and investment due to delays in disbursements by local governments and disbursement under the Emergency Decree on Water Management. Nevertheless, disbursements on budgetary spending by the central government and investments by state-owned enterprises were at normal levels.

The economic outlook for 2013 Q2[3/]

In 2013 Q2, the Thai economy was likely to expand from the same period last year, but at a slower rate compared to the previous quarter, owing mainly to a slowdown in private consumption. At the same time, investment and exports contracted slightly on the back of slower global economic recovery. This was consistent with manufacturing production which was expected to contract due mainly to the decline in export-oriented production. The contraction was also owed to some transitory

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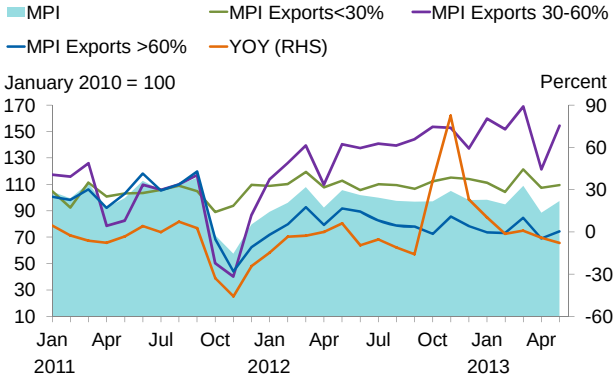
[3/] Economic indicators used in assessing the outlook for 2013 Q2 were produced by the Bank of Thailand, except data on the Manufacturing Production Index and capacity utilization rate, which were provided by the Office of Industrial Economics. The number of tourists and occupancy rate were, in part, compiled by the Tourism Authority of Thailand. Data on the labor market were obtained by the National Statistical Office. The government expenditure data were provided by the Comptroller General's Department, and compiled by the Fiscal Policy Office, Ministry of Finance. The economic outlook was obtained from the Economic/Business Information Exchange Program between the Bank of Thailand and the business sector.

factors, namely temporary supply shortages and a decline in production in some industries as a result of their cooperative efforts to reduce energy usage in April. Meanwhile, the services sector was expected to expand favorably.

Manufacturing production in the second quarter was expected to contract from the same period last year, mainly stemming from the decline in export-oriented production **(Chart 2.8)**. Production of hard disk drives and integrated circuits and parts declined on the back of weak external demand despite its slow recovery. Furthermore, the frozen seafood industry also decreased due to shrimp shortages following the disease outbreak which protracted since the end of last year. Overall production for domestic demand also contracted. However, the contraction was in part a result of lower petroleum production due to closures of refineries for maintenance in the latter part of the quarter. In addition, a number of industries reduced production in April to comply with the energy reduction policy. Nevertheless, the automobile industry was expected to continue to expand, although slowing down somewhat compared to the previous quarter as deliveries of cars under the first-car tax rebate scheme were nearly filled. The decline in overall manufacturing production led to a lower capacity utilization rate than the same period last year **(Chart 2.9).**

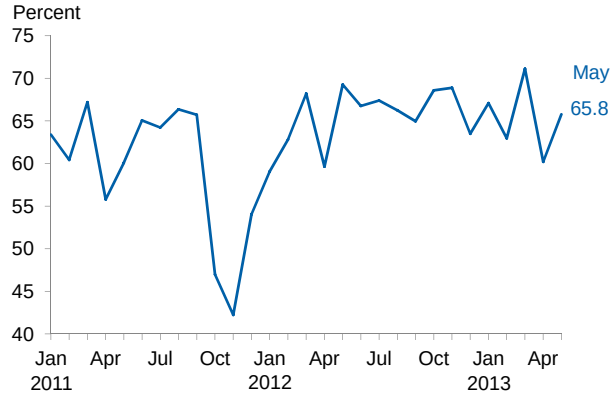
The **services sector** in the second quarter was expected to expand well from the same period last year, despite some deceleration when compared to the previous quarter. Trade and transportation sectors were expected to moderate following a slowdown in economic activity both in private spending and manufacturing production. Meanwhile, the hotel and restaurant sector continued to grow strongly, as reflected by the number of tourists,

Chart 2.8 Manufacturing Production Index (MPI)



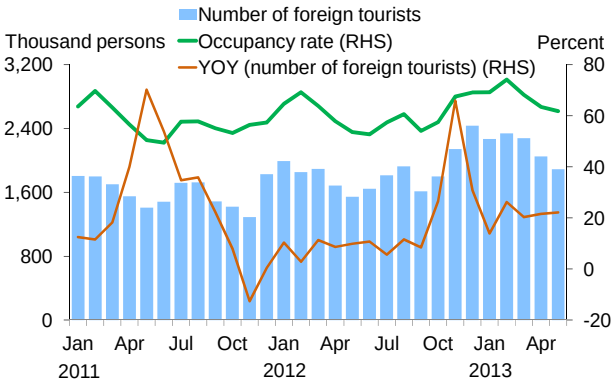
Source: Office of Industrial Economics, Ministry of Industry

Chart 2.9 Capacity utilization rate



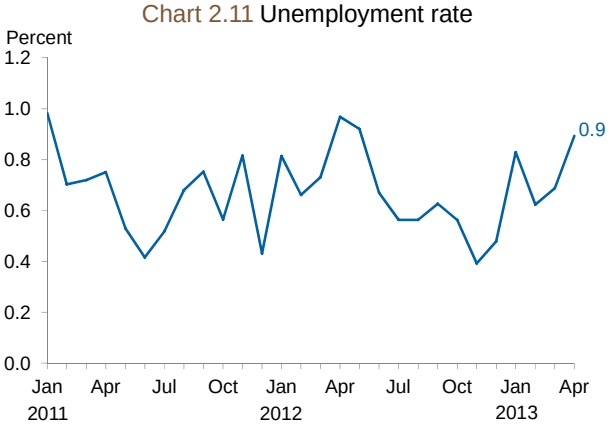
Source: Office of Industrial Economics, Ministry of Industry

Chart 2.10 Number of foreign tourists and hotel occupancy rate



Source: Department of Tourism and Bank of Thailand

Chart 2.11 Unemployment rate



Source: National Statistical Office

mostly from Russia, China and Malaysia, in April and May which continued to increase. This led to a higher hotel occupancy rate when compared to the same period last year **(Chart 2.10)**. The construction sector should expand at a slower rate in line with private investment in construction and constraints from labor shortages and increasing costs.

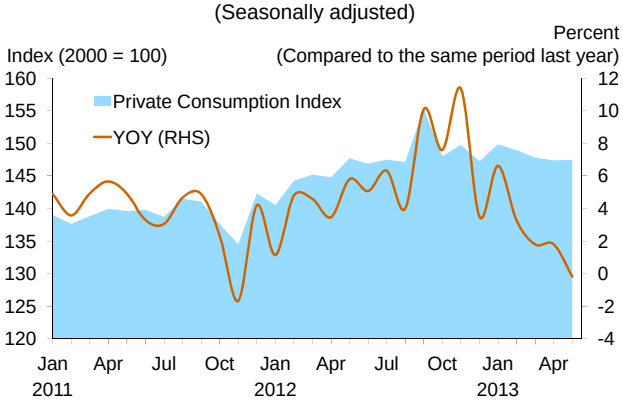
Agricultural production in the second quarter was likely to expand only slightly from the same period last year despite the anticipation of an increase in rubber and palm oil outputs following expansion in the planting area. This was due to a decline in rice output as a result of unfavorable water supply and weather conditions. A high base effect should also be noted as rice output in the same period last year was much higher than usual due to an increase in rice cultivation to compensate for those damaged by the previous year's floods.

Although production was likely to slow down during this quarter, firms retained their workers thanks to the tight **labor market** condition experienced in the past periods. The Business Sentiment Survey also revealed that most firms desired to hire more workers, but was faced with recruiting difficulties. As a result, the unemployment rate in this quarter was likely to be low, similar to the level recorded in the same period last year **(Chart 2.11).** However, a slowdown in production and impacts from the nationwide minimum wage hike to 300 baht per day led entrepreneurs outside the agricultural sector to reduce overtime hours in order to mitigate the impact from rising costs. This was reflected in the declining number of non-farm workers working overtime which was observed since the beginning of the year.

On the demand side, in 2013 Q2, consumption was expected to slow down from the same period last year. Exports were likely to recover slowly following still fragile external demand. Private investment contracted slightly, while public investment continued to lend support to economic growth.

Private consumption was expected to moderate from the same period last year, as reflected by a declining trend of the Private Consumption Index in April and May **(Chart 2.12)**. This was attributable in part to last year's high base when large automobile manufacturers returned to full capacity production last year after the floods, as well as the near complete delivery of automobiles under the first-car tax rebate scheme. In addition, farm income declined in line with prices of rice and rubber. Nevertheless, there remained supporting factors for consumption including high employment and non-farm income, as well as accommodative monetary conditions which would help support spending.

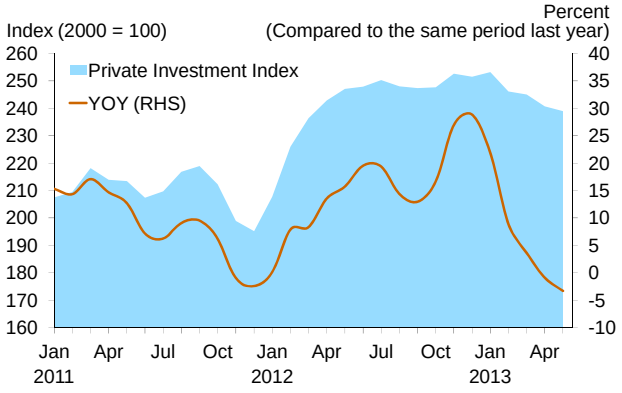
Chart 2.12 Private Consumption Index (PCI)
(Seasonally adjusted)



Source: Bank of Thailand

Private investment was expected to decline from the same period last year, as reflected by the downward trend of the Private Investment Index in April and May **(Chart 2.13)**. This was partly due to a high base effect in the previous year following investment in machinery and equipment after they were promptly imported by manufacturers to replace those damaged by the flood. In addition, the slowing domestic economy combined with fragile external demand reduced the need for immediate investment and led some manufacturers to delay their investment plans. Nonetheless, the private sector still had investment plans in the pipeline. This included investments for manual labor replacement in response to labor shortages and for production process restructuring to enhance efficiency as well as to accommodate anticipated higher domestic and foreign demand following a closer regional economic integration

Chart 2.13 Private Investment Index
(Seasonally adjusted, 3-month moving average)



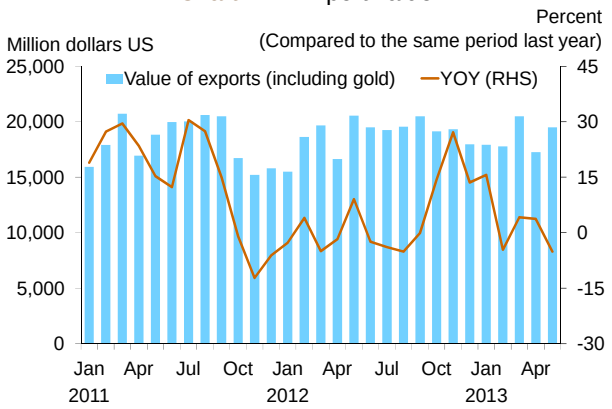
Source: Bank of Thailand

under the ASEAN Economic Community (AEC). This was consistent with the opinions expressed by most entrepreneurs under the Business Liaison Program between the Bank of Thailand and the business sector, and was reflected by the high level of applications and issuance of investment promotion certificates by the Board of Investment, as well as strong business confidence.

Exports of goods were expected to recover slowly from the same period last year as the repercussions from the global economic crisis remained **(Chart 2.14)**. Exports of key manufacturing products in nearly all product categories, with the exception of automobiles, were expected to slow down. Moreover, exports of fishery and agricultural products were likely to moderate due to shrimp supply shortages for frozen seafood production and a decline in rice exports following deteriorating price competitiveness. Nevertheless, **exports of services** were expected to continue improving from the previous quarter, prompted by an increase in foreign tourists from almost all regions, in particular those from Russia and China.

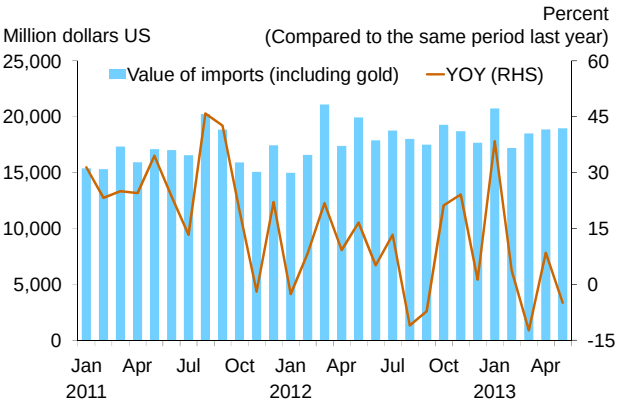
Imports of goods were likely to decelerate from the same period last year **(Chart 2.15)** stemming from a slowdown in capital goods imports following the private investment trend. In addition, imports of raw materials and intermediate goods moderated in line with manufacturing production, while imports of consumer goods were likely to slow in line with private spending. Meanwhile, **imports of services** were expected to expand at a rate similar to that in the previous quarter, due to travel expenditures which increased in tandem with the number of Thais travelling abroad. However, freight payments declined in tandem with import volume.

Chart 2.14 Export value



Source: Bank of Thailand

Chart 2.15 Value of imports

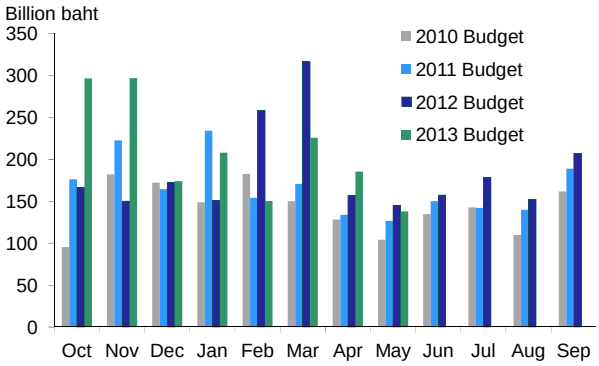


Source: Bank of Thailand

Fiscal stimulus was likely to increase from the same period last year. This was a result of continuous budget disbursements which were consistent with the government's disbursement plan **(Chart 2.16)**. Furthermore, local government expenditures were likely to increase following large budget allocations in the previous periods. Nevertheless, non-budgetary expenditures remained low due to a delay in the contractor selection process for the water management projects.

In summary, the Thai economy in 2013 Q2 was expected to moderate from the previous quarter owing both to tepid domestic demand and exports, consistent with the slowdown in manufacturing production.

Chart 2.16 Disbursement of government budget



Note: Excluding principal repayments and replenishments of the treasury reserve
Source: Comptroller General's Department, Ministry of Finance

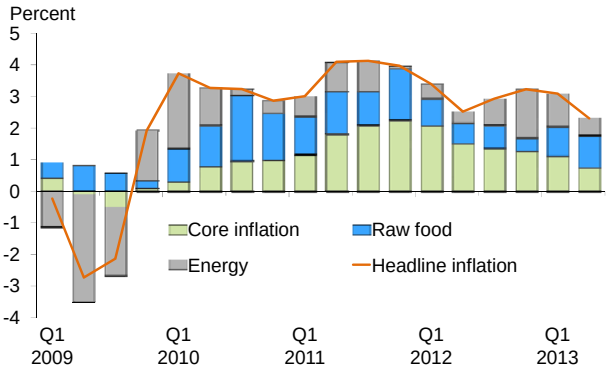
2.3 Costs and prices

Inflation

Headline inflation in 2013 Q2 stood at 2.32 percent, declining from 3.09 percent in the previous quarter **(Chart 2.17)** due to a slowdown in energy prices and core inflation. Energy prices in 2013 Q2 increased year-on-year by merely 4.60 percent, decelerating considerably from 8.76 percent observed in the previous quarter. The slowdown was prompted by a fall in domestic retail oil prices following world crude oil prices which was on a decline in accordance with the fragile global economic recovery.

Meanwhile, prices of raw food rose year-on-year by 6.60 percent, increasing slightly from 6.08 percent in the previous quarter due to an increase in prices of vegetables and fruits, meat, poultry, aquatic animals and eggs following extremely hot weather conditions which led to low outputs.

Chart 2.17 Contribution to headline inflation



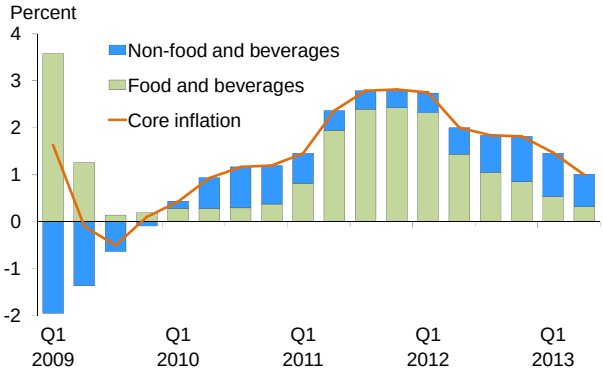
Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand

Table 2.2 Quarterly inflation

Unit: Percent	2012	2012			2013	
		Q2	Q3	Q4	Q1	Q2
Percentage change from previous year (%Δyoy)						
- Headline Consumer Price Index	3.02	2.52	2.94	3.23	3.09	2.32
• Core Consumer Price Index)	2.09	2.00	1.84	1.82	1.47	1.00
• Raw food	4.40	4.21	4.80	2.92	6.08	6.60
• Energy	7.09	3.02	7.54	14.79	8.76	4.60
Percentage change from previous quarter (%Δqoq)						
- Headline Consumer Price Index	-	1.2	1.0	0.4	0.5	0.5
• Core Consumer Price Index	-	0.6	0.4	0.3	0.2	0.1
• Raw food	-	3.1	2.2	0.3	0.4	3.6
• Energy	-	2.4	2.6	1.3	2.1	-1.5
Percentage change from previous quarter (%Δqoq_sa)						
- Headline Consumer Price Index	-	0.6	1.0	1.0	0.4	-0.1
• Core Consumer Price Index	-	0.3	0.5	0.5	0.2	-0.1
• Raw food	-	1.1	2.2	1.0	1.7	1.7
• Energy	-	0.8	3.9	4.6	-1.0	-2.4

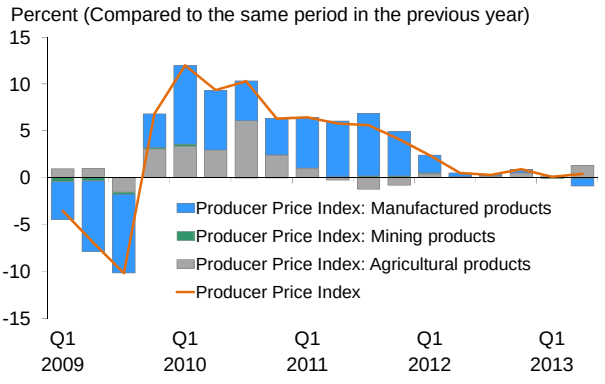
Source: Trade and Economic Index Bureau, Ministry of Commerce, and calculations by Bank of Thailand

Chart 2.18 Contribution to core inflation



Source: Trade and Economic Index Bureau, Ministry of Commerce, and calculations by Bank of Thailand

Chart 2.19 Contribution to Producer Price Index



Source: Trade and Economic Index Bureau, Ministry of Commerce and calculations by Bank of Thailand

Core inflation in 2013 Q2 stood at 1.00 percent, slowing down continuously from 1.47 percent in the previous quarter **(Chart 2.18)** owing mainly to low production costs, as well as sluggish domestic demand and high market competition. This resulted in limited pass-through of costs to prices of goods and services, as reflected by a contraction in the Core Consumer Price Index on a quarter-on-quarter, seasonally adjusted basis **(Table 2.2).**

Production cost conditions

Costs of production remained low overall, as reflected by an increase in the Producer Price Index of 0.4 percent in 2013 Q2, a slight increase from 0.1 percent in the previous quarter **(Chart 2.19).** This followed the acceleration in the price of agricultural products, mainly vegetables and fruits. However, the price of manufactured products contracted further owing to a decline in the price of petroleum products in line with world crude oil prices. At the same time, the price of rubber and plastic products decreased following a continuous decline in purchases by China, the world's largest buyer of rubber.

Inflationary pressure in the periods ahead

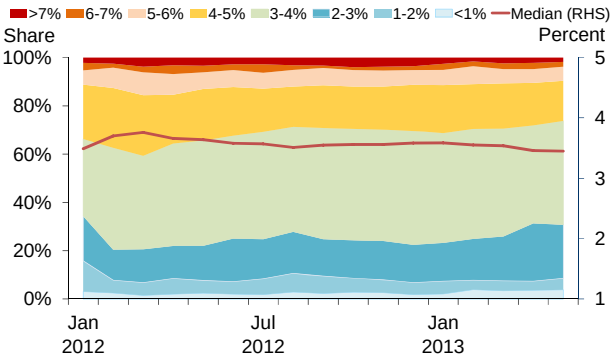
Inflationary pressure in the short-term was likely to rise from the current level but would remain low, mainly due to the slight increase in domestic demand pressure, while costs and inflation expectations remained stable.

(1) Demand pressure was likely to increase slightly following gradual improvements in domestic demand. At the same time, exports were expected to gradually recover on the back of trading partners' economic recovery. Overall, the output gap remained close to zero in 2013 Q4, reflecting low inflationary pressure.

(2) Going forward, cost pressure was likely to remain stable compared to the current level, in tandem with world prices of crude oil and commodities. The weaker baht would not have much effect on domestic energy cost pressure as the government was expected to maintain domestic retail diesel prices. In addition, upward adjustment to the LPG price, which would take effect in September 2013, was expected to have minimal impact on cost pressure as the price adjustment would be done at a gradual pace.

(3) Inflation expectations remained stable, as reflected by the 12-month-ahead inflation expectations in May 2013 which remained close to that of the previous month at 3.5 percent **(Chart 2.20).**

Chart 2.20 12-month ahead inflation expectation
(May 2013)



Source: Bank of Thailand's Business Sentiment Survey

Labor constraints and economic growth

.During the past several years, the Thai labor market has been tightening, with unemployment rate standing at 0.9 percent in April 2013, the lowest level amongst Asian countries. Tight labor market has been one of the key structural constraints facing the Thai economy, and an important impediment to businesses in Thailand. This is in line with business survey results which found that labor recruitment has become increasingly difficult in the past 2-3 years. Labor supply constraint is thus an often-cited problem among businesses. However, another important issue of the Thai labor market that should also be addressed concurrently is the slow productivity growth of Thai labor.

Impact on Thailand's economic growth

The Thai economy has been driven by both employment and productivity growth **(Chart 1)**. However, since the Asian financial crisis in 1997, Thai economic growth slowed down considerably as a result of lower growth in both the employment and productivity.

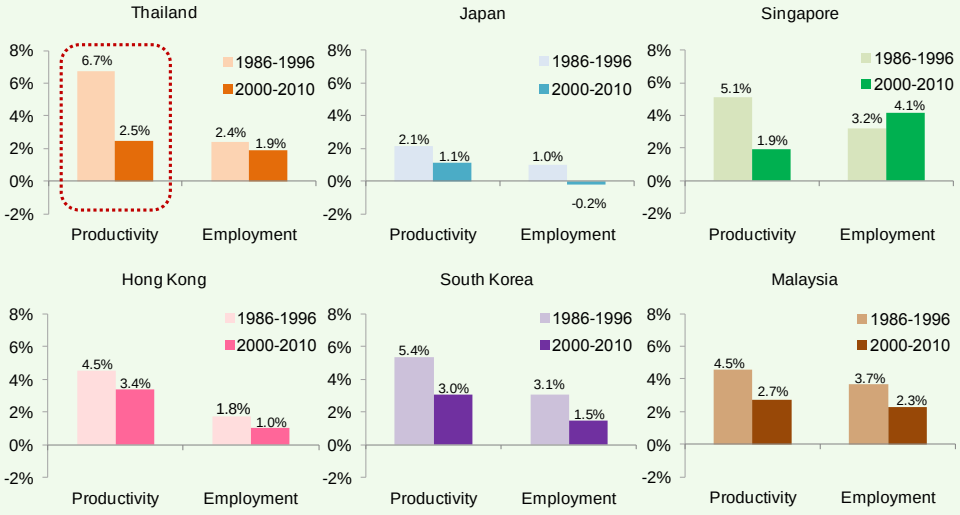
Chart 1 Economic growth



Source: Labor force survey, Office of the National Economic and Social Development Board, calculated by Bank of Thailand

With regard to low employment growth, during 2000-2010, the employment grew on average by 1.9 percent per year, compared to 2.4 percent per year during 1986-1996. This was mainly due to a continuous drop in the fertility rate. Most recently, the average number of children born to one female during her reproductive years is 1.6 children, resulting in low working-age population growth. Furthermore, Thailand is one of the first nations to become an aging society in ASEAN. Given these factors together with the fact that Thailand's labor participation rate is already quite high, standing at 71 percent of total labor force in 2012, further labor force expansion will likely be difficult.

Chart 2 Productivity of Thai labor compared to other countries



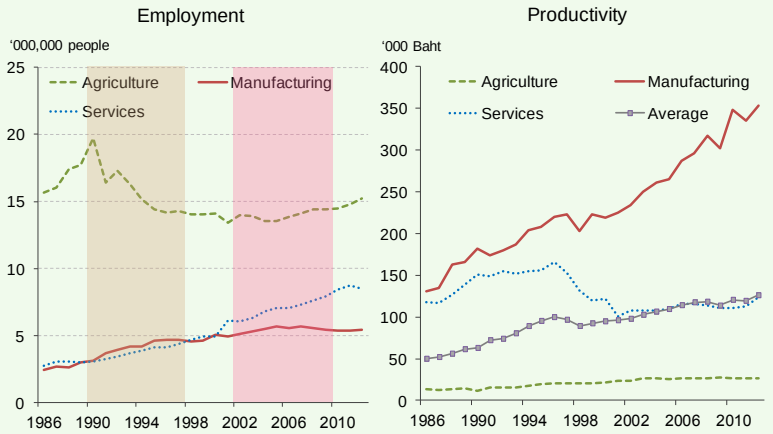
Source: Labor force survey, Office of the National Economic and Social Development Board, World Bank, International Monetary Fund, calculated by the Bank of Thailand

With regard to low productivity growth, the ratio of productivity growth to employment growth fell from 2.86 prior to the 1997 crisis to a mere 1.56 in the period after the crisis. This shows that Thai labor productivity has slowed down by more than the rate of employment growth slowdown, which is the main factor weighing on economic growth. In addition, when compared to other countries **(Chart 2)**, it was found that in the decade following the Asian crisis, economic growth from labor productivity and employment slowed in nearly every country. Interestingly, however, the slowdown in employment was not much different among these countries (with the exception of Singapore where employment accelerated from the government's policy), while the rate of deceleration in productivity growth differed, with the sharpest slowdown in the case of Thailand.

What accounts for slow productivity growth

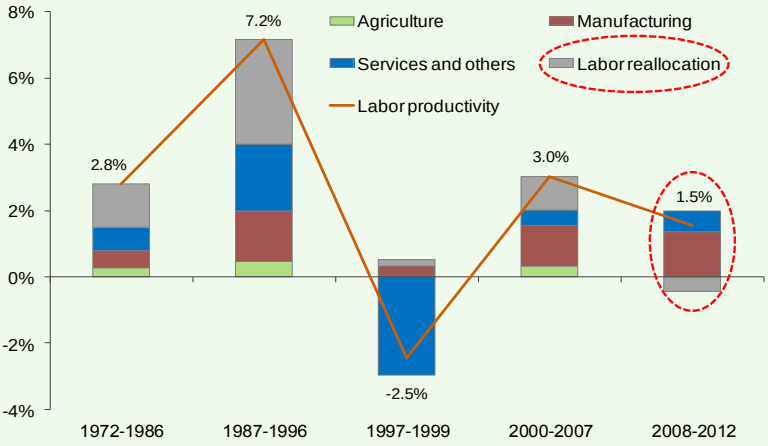
Labor productivity in Thailand slowed down considerably due to (1) labor reallocation from high productivity sectors to low productivity sectors.[1/] In the period leading up to the Asian crisis, Thai labor continuously moved away from the agricultural sector, which had low productivity, to the manufacturing and services sectors, which had higher productivity. Employment in the agricultural sector declined from approximately 20 million in 1990 to approximately 14 million in 2000 **(Chart 3)**. As a result, this pattern of labor reallocation had contributed as a key driver of productivity growth in Thailand **(Chart 4)**. However, more recently, this driver became weakened and eventually reversed its direction. Its contribution to productivity growth turned negative, particularly during 2011-2012 when up to 700,000 people moved into

Chart 3 Thai employment by sector



Source: Labor force survey, Office of the National Economic and Social Development Board, calculated by the Bank of Thailand

Chart 4 Contribution to labor productivity growth



Source: Labor force survey, Office of the National Economic and Social Development Board, Sra Chuenchoksan and Don Nakornthab, 2008, calculated by the Bank of Thailand

[1/] Following the calculation used in the study by Sra Chuenchoksan and Don Nakornthab, 2008, and Nicholas Oulton and Sylaja Srinivasan, 2005, which calculated the weighted average of each sector in the economy and accounted for the effect of labor reallocation on overall productivity.

the agricultural sector. This was partly due to the incentive arising from rising agricultural prices in both domestic and global markets. (2) Slower capital accumulation **(Table 1)**. In 2008-2011, the growth rate of total capital to labor ratio fell to a mere 1.4 percent compared to 8.5 percent before the Asian crisis. In part, this was because many industries that relied heavily on labor were able to continue their businesses on the back of unskilled migrant labor supply from neighboring countries. These industries therefore lacked the motivation to invest, for example, to replace manual labor with machineries or to move up the value chain to a more complex and more value-added production. Low investment in terms of both quantity and quality led to a significant slowdown in productivity growth from capital deepening, which declined from 4.3 percent in 1987-1996 to 1.2 percent in 2009-2011.

Table 1 Rate of expansion of capital to labor ratio

1987-1996	1997-1999	2000-2007	2008-2011
8.5%	3.9%	0.6%	1.4%

Source: Labor force survey, Office of the National Economic and Social Development Board, calculated by the Bank of Thailand

In conclusion, labor constraint problems arose from both the shortage of labor and a slowdown in labor productivity. This led to a decline in Thailand's potential economic growth in recent years, and the problems could intensify going forward. Resolution to this structural problem will take time and must take into consideration the costs and benefits in the short and long term. Short-term measures to address this problem, such as expanding the number of immigrant labors so as to reduce constraints in the manufacturing sector in the short run, without controlling for the type of work allowed, could in turn reduce businesses' motivation to invest to increase their potential in the long term. More importantly, measures that will encourage reallocation of existing labor to highly productive sectors should be fostered. This can be achieved by removing measures that may distort returns in each economic sector, while implementing measures that promote continuous restructuring of production as well as investing in public infrastructure to enhance private investment capacity.

Monetary and Financial Stability

3. Monetary and Financial Stability

Thailand's overall economic and financial stability continued to be well maintained. Money market interest rates adjusted downwards in tandem with the policy interest rate. Meanwhile, stability in financial institutions and corporate sectors remained intact, while signs of over-expansion in the real estate sector began to dissipate.

Nonetheless, financial markets and households became more fragile. The Stock Exchange of Thailand Index and the Thai baht became significantly more volatile during this period following the trend in global markets which was driven in part by market anticipation of a sooner-than-expected tapering of the US quantitative easing measures. With regard to the household sector, debt burden continued to rise as a result of the strong pace of household debt expansion in the previous period. As for the banking sector, high competition for deposits between commercial banks and Specialized Financial Institutions (SFIs) in anticipation of loan demand expansion and the funding needs related to the government's large investment projects could negatively impact banks' liquidity in the next periods.

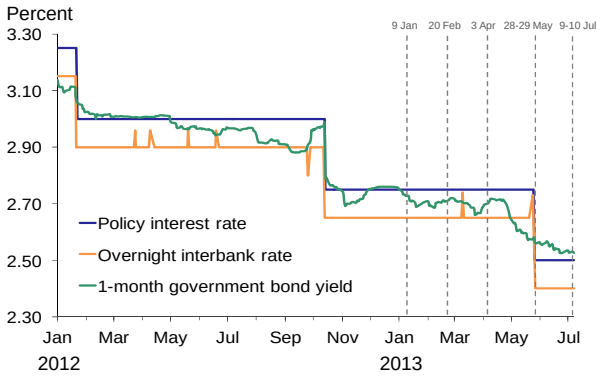
3.1 Financial markets

Money market rates adjusted downwards in line with the policy interest rate. Government bond yields were somewhat volatile owing to foreign investors' sales as well as market expectation of a decrease in the policy interest rate at the May 29, 2013 Monetary Policy Committee (MPC) meeting.

Money and bond markets

In April and May of this year, short-term money market rates remained stable and close to the policy interest rate, before adjusting downwards in line with the MPC's decision to cut the policy interest rate from 2.75 to 2.50 percent per annum at the May 29, 2913 meeting. The overnight interbank lending

Chart 3.1 Money market short-term interest rates

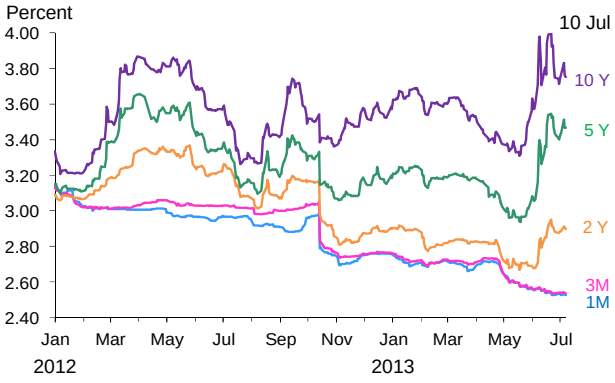


Sources: Bank of Thailand and the Thai Bond Market Association (ThaiBMA)

rate stood at 2.40 percent at the end of the second quarter, while the one-month government bond yield trended downwards throughout the quarter, following the policy rate cut, to reach 2.53 percent at the end of June. Subsequently, at the July 9-10, 2013 meeting, the MPC decided to maintain the policy interest rate at 2.50 percent per annum, causing short-term interest rates on July 10, 2013 to remain close to the 2013 Q2 end level **(Chart 3.1)**.

Short-term government bond yields in 2013 Q2 edged lower in line with the policy interest rate and an increase in demand by both domestic and foreign investors. Moreover, medium to long-term government bond yields trended down throughout April and the first half of May along with movements in the US government bond market which continued to be dictated by fragile confidence in the global financial markets, foreign investors' portfolio rebalancing to longer-term government bonds, and market expectation of a decrease in the policy interest rate at the May 29, 2013 MPC meeting. For the rest of the quarter, however, medium to long-term government bond yields rose significantly in tandem with an increase in the US government bond yield, owing to foreign investors' sales that were driven by an anticipation of a sooner-than-expected tapering of the US quantitative easing measures. The shift in risk appetite also led to an increase in volatility during this period **(Chart 3.2)**.

Chart 3.2 Government bond yields



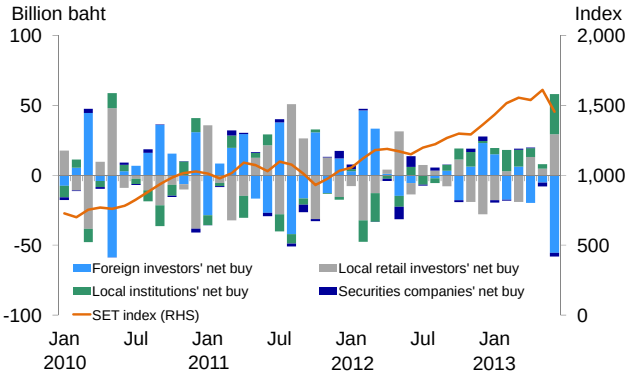
Sources: The Thai Bond Market Association (ThaiBMA)

Equity market

Volatility in the Stock Exchange of Thailand (SET) Index in 2013 Q2 rose significantly. The SET index was seen to rise, peaking at 1,643.43 on May 21, 2013, the highest level in 19 years, with a notable increase in retail investors' holdings. However, for the rest of the quarter, the SET index declined to stand at 1,364.09 on June 24, 2013. This was due mainly to the continued sell-offs by foreign investors

since late April 2013 triggered by the anticipation of a sooner-than-expected unwinding of the US quantitative easing measure which some viewed that it could adversely affect global liquidity. Other factors that caused the SET index to decline included investors' rebasing of the SET index and short-term profit taking sales after the index had accelerated sharply in the previous period. In this light, accumulated net sales by foreign investors from April to June of this year totaled 80 billion baht[1/] **(Chart 3.3)**.

Chart 3.3 The Stock Exchange of Thailand index
(up to June 28, 2013)



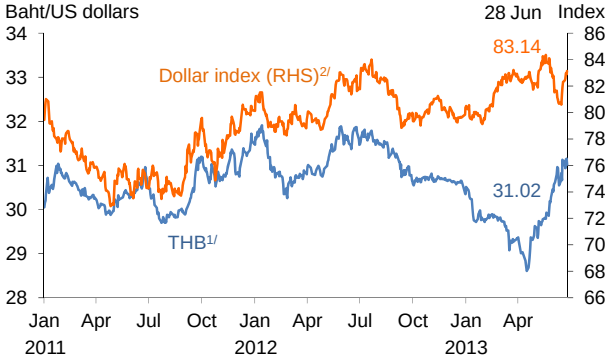
Source: The Stock Exchange of Thailand

Key risks to be monitored in the periods ahead included (1) the outlook of major economies, in particular the US, and the decelerating growth trend of the Chinese economy, which could result in capital outflows from emerging market economies, and (2) the outlook of Thailand's economy during the latter half of the year, which could affect volatility of the SET index.

Foreign exchange market

At the end of 2013 Q2, the Thai baht stood at 31.02 baht per US dollar, depreciating by 5.91 percent from the end of the previous quarter (Chart 3.4). The volatility of the Thai baht increased markedly during this period, as at the beginning of the quarter the Thai baht was still on its appreciating trend that began since the start of the year. The appreciation during the earlier period was attributable to several factors including (1) Thailand's strong economic fundamentals relative to those of regional and advanced economies, (2) an increase in capital inflows partly led by an

Chart 3.4 Exchange rate
and trade-weighted dollar index



Sources: 1/ Bank of Thailand
2/ Bloomberg

1/ On June 26, 2013, foreign investors' net sales totaled 16.9 billion baht, due in part to sales of SHV Nederland B.V.'s MAKRO of 17.6 billion baht, which was part of the acquisition of Siam Makro Public Company Limited by CP ALL Public Company Limited.

expectation of higher global liquidity following Japan's announcement of additional stimulus measures, and (3) exporters' sales of US dollars to hedge against foreign exchange risks. Together, these factors moved the Thai baht to peak at 28.61 baht per US dollar on April 19, 2013.

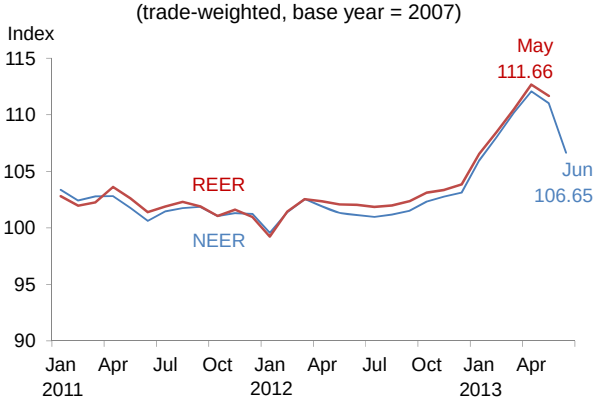
Towards the end of April, however, the Thai baht began to reverse its direction. The depreciation that followed could be viewed in part as a market correction after the currency had appreciated substantially since the beginning of the year. This was reinforced by investors' anxiety regarding a possible implementation of capital flow management measures to curb the rapid appreciation. In addition, the lower-than-expected GDP in 2013 Q1 also raised concerns about the Thai economic outlook and possible credit rating downgrade, which subsequently caused foreign portfolio investments to slow down.

From the end of May to the start of June, the Thai baht depreciated sharply as a result of investors' anticipation of a sooner-than-expected unwinding of the US quantitative easing measures, given clearer signs of US economic recovery. Capital flows began to pour out of emerging market economies and into the US and other advanced economies. As such, the Thai baht along with regional currencies depreciated sharply and continued trending downwards through the end of the quarter. In the second quarter, the Thai baht averaged at 29.90 baht per US dollar and stood at 31.02 baht per US dollar at the end of the quarter, depreciating by 8.42 percent from the strongest level recorded on April 19, 2013.

The **Nominal Effective Exchange Rate (NEER)** averaged at 109.91 in 2013 Q2, depreciating by 1.73 percent from the previous quarter's average. In mid-April, the NEER rose to its strongest level due

mostly to the substantial weakening of the Japanese yen. Afterwards, the NEER started to depreciate in line with regional economies owing to the strengthening of major currencies, namely, the US dollar, the euro, as well as the Chinese renminbi, following an improvement in the global economic outlook. Meanwhile, the **Real Effective Exchange Rate (REER)** averaged at 111.66 in May, appreciating by 1.08 percent from the end of the previous quarter, in line with the NEER during the same period **(Chart 3.5)**.

Chart 3.5 NEER and REER
(trade-weighted, base year = 2007)



Source: Bank of Thailand

3.2 Financial institutions

In 2013 Q2, deposit and lending rates of the four largest commercial banks remained unchanged from the previous quarter partly due to high competition for deposits between commercial banks and SFIs. This was reflected in part by continued releases of special deposit campaigns paying higher interest rates than those of normal time deposits in order to meet loan demand and the funding needs related to the government's large-scale investment projects. As a result, deposits of financial institutions accelerated from the previous quarter. Meanwhile, private credits slowed down slightly, both in the consumer and corporate loan components.

Interest rates, credits and deposits

Commercial banks' deposit rates in 2013 Q2 decreased in line with the policy interest rate with small- and medium-sized banks responsible for most of the move. At the end of the quarter, interest rates on the 12-month deposits of small and medium sized banks averaged at 2.68 percent per annum, declining from 2.74 percent per annum in the previous quarter. A number of commercial banks,

however, started to mobilize deposits through special deposit products which paid higher interest rates than those of normal time deposits.

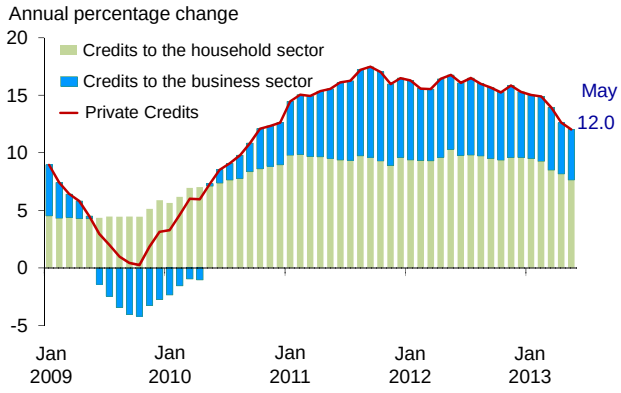
Interest rates on the 12-month deposits of the four largest commercial banks[2/], however, remained unchanged from the previous quarter, standing at 2.40 percent per annum, owing to high competition for deposits in anticipation of private loan demand expansion and the funding needs under the government's large-scale investment projects.

Meanwhile, the average MLR of the four largest commercial banks remained unchanged from the previous quarter at 7.00 percent per annum.

Private credits at end-May 2013 grew at 12.0 percent from the same period last year, decelerating from 14.0 percent in the previous quarter **(Chart 3.6)**. The slowdown in private credits was attributed to both household and corporate loans. Although consumer loans, especially auto leasings and other personal loans[3/], expanded at a high level, it was likely to slow down as consumers became more cautious in their spending behavior after having paid large sums previously to take advantage of the first-car tax rebate scheme. Meanwhile, corporate loans slowed down in line with private investment.

Deposits and bills of exchange (B/E) at end-May 2013 expanded at 11.9 percent from the same period last year, accelerating from 9.6 percent in the pervious quarter. The increase in deposits was induced in part by commercial banks' and SFIs' continued releases of special deposit campaigns that paid attractive interest rates in order to mobilize

Chart 3.6 Other Depository Corporations' private credits



Source: Bank of Thailand

[2/] Bangkok Bank, Krungthai Bank, Kasikorn Bank and Siam Commercial Bank

[3/] Consists of personal loans under BOT's supervision (personal loans without collateral) and other personal loans that are outside BOT's supervision

funds to meet private loan demand and the public funding needs related to the government's large-scale infrastructure investment projects. The fierce competition for deposits could tighten commercial banks' liquidity in the periods ahead.

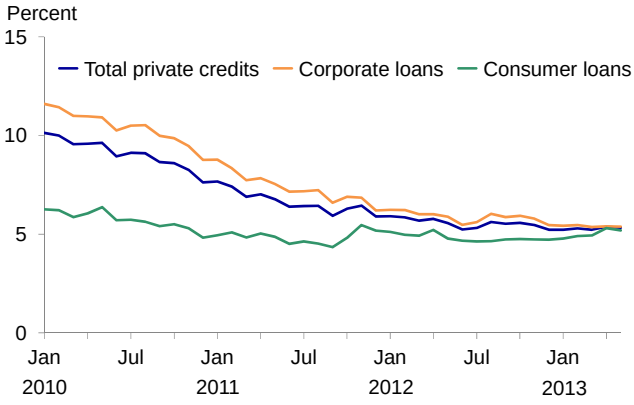
Stability of financial institutions

The overall stability of financial institutions in 2013 Q2 remained sound as reflected by strong profitability, high capital adequacy ratio and low delinquency and NPL ratio.

Commercial banks continued to pose strong profitability in 2013 Q1 whereby net profit stood at 52.3 billion baht, increasing from 39.2 billion baht recorded in the previous quarter **(Table 3.1)**. Moreover, the ratio of capital adequacy remained high. The implementation of Basel III at the beginning of the year only caused a small impact on commercial banks' capital adequacy ratio. In this regard, the ratio of capital adequacy stood at 16.1 percent in April 2013, increasing from 15.9 percent in March. With regard to liquidity, the loan to deposit and B/E ratio edged up from 93.1 percent at the end of 2012 Q4 to 93.4 percent at the end of 2013 Q1, signaling a slight tightening in commercial banks' liquidity. Nonetheless, loan quality in the banking system remained sound as reflected by the delinquency and NPL ratios of private credits which continued to stabilize at a low level of 5.3 percent **(Chart 3.7)**.

Looking ahead, loan quality would need to be monitored closely given the potential impact of the Thai economic slowdown on corporates' and households' debt servicing abilities.

Chart 3.7 Delinquency and NPL ratio



Source: Bank of Thailand

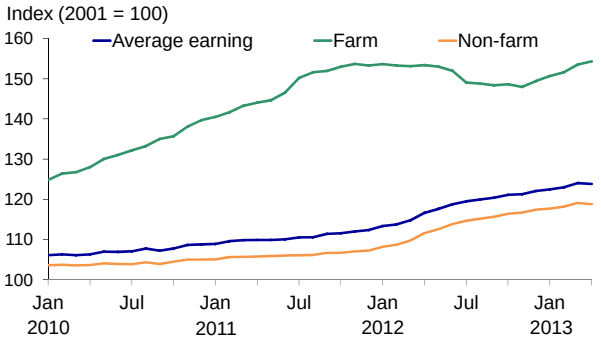
3.3 Non-financial sectors

Stability in non-financial sectors remained intact overall. Household income and employment were high, while corporate enjoyed strong profitability. However, key risks that warranted monitoring in the next periods included the weakening of economic outlook which could affect debt servicing ability of households as well as corporate performance.

Household sector

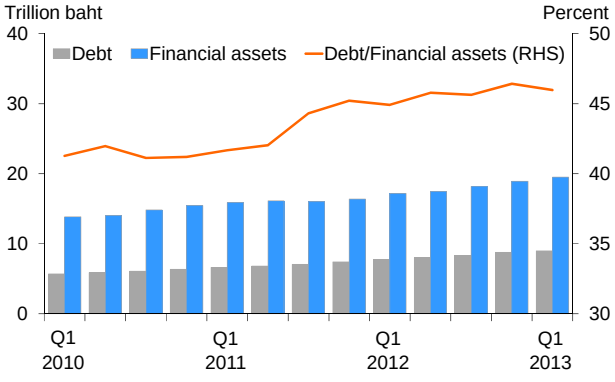
Stability of the household sector remained sound, with high levels of employment and household income. However, households' debt servicing ability would still require close monitoring due to rising debt burden.

Chart 3.8 Households' income*



Note: *seasonally adjusted, real income (12-month moving average)
Sources: Office of Agricultural Economics, National Statistics Office, calculations by Bank of Thailand

Chart 3.9 Ratio of debt to financial assets



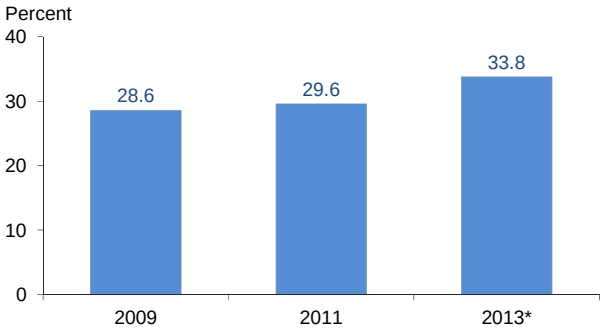
Source: Bank of Thailand

Households' income and employment remained high, with rising average income in both farm and non-farm sectors **(Chart 3.8)**. Employment conditions remained favorable, as reflected by the low unemployment rate of 0.9 percent in April 2013. Moreover, debt servicing ability of households remained satisfactory, as reflected by a low NPL ratio which edged up only marginally from 4.7 percent at the end of 2012 to 4.9 percent at the end of 2013 Q1. Moreover, households' liquidity improved slightly, as reflected by a decline in the ratio of debt to financial assets from 46.4 percent at the end of 2012 to 46.0 percent at the end of 2013 Q1 **(Chart 3.9)**.

Nonetheless, the sharp acceleration in consumer loan growth in previous periods led to an increase in households' debt burden. Information from the Household Socio Economic Survey of 2013 Q1 that was conducted by the National Statistical Office (calculated by the Bank of Thailand) showed that the overall ratio of households' average monthly

debt payments to income was 33.8 percent, increasing from 29.6 percent in 2011 **(Chart 3.10).** In this light, households' debt servicing ability, especially for low income households, warranted close monitoring as the situation could become deteriorate should the Thai economy weaken further.

Chart 3.10 Debt Service Ratio (DSR)



Note: *2013 Q1 survey data
Sources: NSO's households' socio-economic survey and calculations by Bank of Thailand

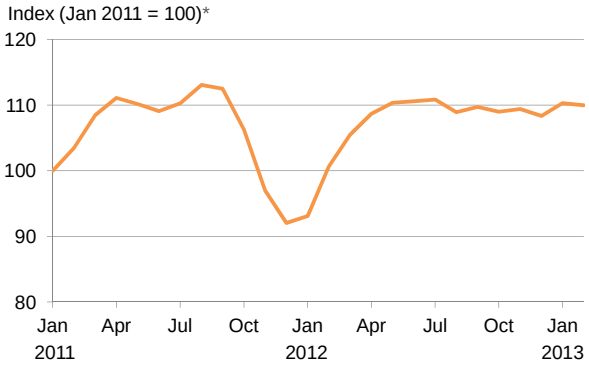
Corporate sector

Corporate profitability in 2013 Q1 improved from the previous quarter regardless of the impact of the slowdown of global as well as domestic demand which had led to persisting decline in manufacturing sales. Lower costs of raw material imports and a significant improvement in the hotel and restaurant sector were among key factors that contributed positively to the overall corporate profitability.

Corporate profitability in 2013 Q1 increased, especially in the manufacturing sector where profit started to improve even though manufacturing sales were still on a decline due to the slow global recovery **(Chart 3.11)** and the softening of domestic demand. In this regard, lower costs of raw material imports, owing partly to the decrease in world crude oil price as well as the stronger Thai baht, lent support to favorable firm profitability. The overall operating profit margin (OPM) increased from 5.8 percent in the previous quarter to 6.3 percent this quarter. Moreover, the hotel and restaurant sector saw a significant improvement from the previous quarter owing to the record high number of overseas tourists **(Chart 3.12)**. However, the real estate sector which expanded strongly in 2012 Q4 started to slow down as firms experienced labor shortages and limitations in finding suitable locations for new condominium projects. As a result, profitability of the real estate sector declined this quarter.

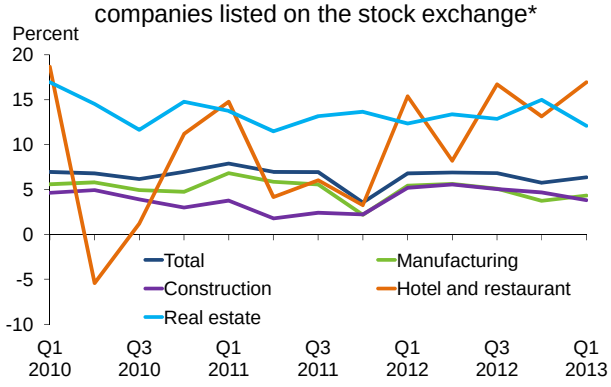
For the next periods, risk factors that must be monitored included uncertainty of the global economy, namely the fragile European economic recovery and the slowdown of the Chinese economy,

Chart 3.11 Thailand's export value index



Note: *Seasonally adjusted, 3-month moving average
Sources: Customs Department, calculations by Bank of Thailand

Chart 3.12 Operating profit margin of non-financial companies listed on the stock exchange*



Note: *Median
Sources: Stock Exchange of Thailand and calculations by Bank of Thailand

as well as new developments regarding the US' quantitative easing measures, and the Thai baht volatility which could potentially affect business operations, particularly export-oriented businesses and businesses with high foreign debt.

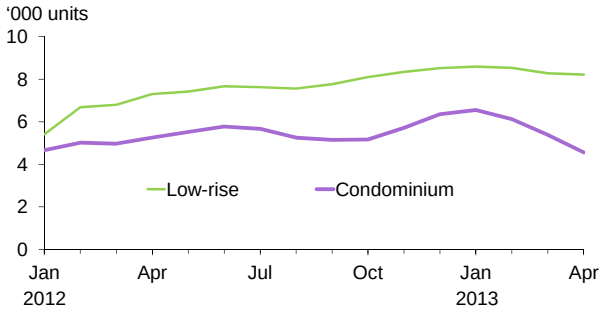
Real estate sector

The real estate sector in 2013 Q2 slowed down from the previous quarter, particularly in condominiums. Nonetheless, overall stability in the real estate sector remained intact. The problem of labor shortages was a main risk factor to monitor as it could weigh importantly on future developments of the real estate sector.

During 2013 Q2, the overall real estate sector slowed down from the previous quarter in both demand and supply. The slowdown in demand was reflected by the number of housing registrations in Bangkok and its vicinity, in particular condominiums, that decreased continuously from last quarter **(Chart 3.13)**. On the supply side, real estate firms began to postpone new openings partly as a result of labor shortages in the building and construction sector. In addition, finding suitable site for condominium development was becoming more difficult. As a result, new openings of condominiums in May 2013 declined. Meanwhile, openings of low-rise residential units continued to increase after the waiting period to assess the flood situation in the previous year ended **(Chart 3.14)**.

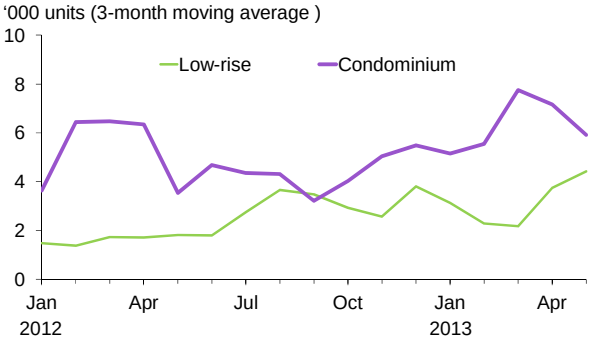
Regarding debt-servicing ability pertaining to the real estate sector, although housing prices continued to rise, household income had also been increasing. This resulted in the ratio of housing prices to income of mortgage borrowers to remain stable at 3.4, close to the past five-year average of 3.2. Moreover, in 2013 Q2, commercial banks began to tighten their credit standards for housing loans after some easing in the first quarter of the year. This

Chart 3.13 Number of housing registrations in Bangkok and its vicinity*



Note: *3-month moving average (data as of April 2013)
Number of housing registrations, including both new and pre-owned units, is used to reflect housing demand
Source: REIC

Chart 3.14 Number of new launches in Bangkok and its vicinity*

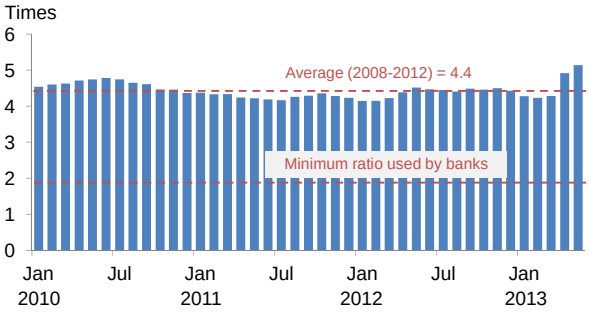


Note: *3-month moving average (seasonally adjusted data as of May 2013)
Source: Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

was reflected by the debt service coverage ratio (DSCR) in May of 5.1, which was higher than the previous five-year's average of 4.4 **(Chart 3.15)**.

Overall, stability remained sound, and signs of overheating in real estate markets in major provinces began to dissipate. In addition, a survey of real estate firms revealed that a majority of firms had in place protection measures against speculation such as charging fees for name changes on the preemption certificate and imposing limits for the number of rooms a customer can buy in a popular project. Nonetheless, speculative activities were still present in some areas, and this issue must be continuously monitored along with the impact of labor shortage problem on the real estate sector going forward.

Chart 3.15 Debt Service Coverage Ratio (DSCR)



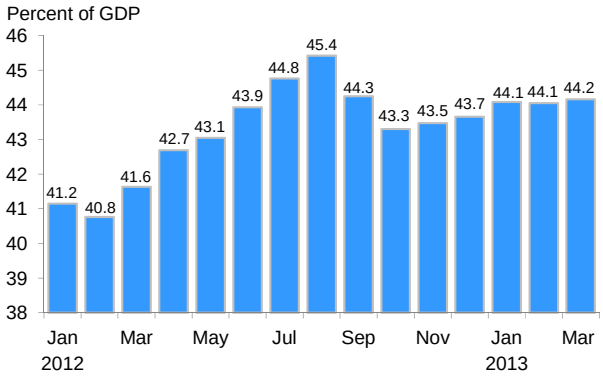
Note:
(1) DSCR = monthly income/monthly debt payment (principal and interest)
(2) Calculated for new mortgages in Bangkok and its vicinity
(3) 50th percentile (3-month moving average)
Source: Agency for Real Estate Affairs (AREA) and calculations by Bank of Thailand

Fiscal sector

Financial position of the fiscal sector remained sound. The government's spending plan under the water management projects would continue to be monitored as it could be further delayed. Fiscal liabilities were likely to increase in the next periods.

Stability of the fiscal sector was well maintained overall, as reflected by the ratio of public debt to GDP at the end of March 2013 of 44.2 percent, which remained close to the previous quarter's level **(Chart 3.16)**. Nonetheless, the government's spending plan, in particular the water management projects could be further delayed after the Administrative Court ruled that a public hearing must be sought in order to proceed, and hence warranted close monitoring. Fiscal liabilities could also increase in the next periods stemming from government spending, in particular from the rice pledging scheme, as well as borrowings to finance infrastructure development of the transportation network.

Chart 3.16 Public debt to GDP



Note: Fiscal year
Source: Public Debt Management Office

Table 3.1 Sectoral Indicators for assessing risks and vulnerabilities to financial stability

Indicators	2010	2011	2012		2013				
			Q3	Q4	Q1	Q2	Apr	May	Jun
1. Financial markets sector									
Bond market									
Bond spread (10 years–2 years)	1.3	0.4	0.3	0.7	0.8	n.a.	0.6	0.7	n.a.
Equity market									
SET Index	849.8	1,025.5	1,228.0	1,314.8	1,500.9	n.a.	1,539.1	1,611.4	n.a.
Actual volatility (SET Index)[1/]	17.6	21.3	11.0	8.9	11.7	n.a.	20.0	15.6	n.a.
Price to earnings ratio (times)	11.5	13.2	17.7	17.3	17.6	n.a.	17.5	18.4	n.a.
FX market									
Actual volatility (baht)	3.4	4.8	4.7	3.3	3.8	n.a.	5.4	6.7	n.a.
Nominal effective exchange rate (NEER)	102.9	101.8	101.2	102.7	108.4	n.a.	112.1	111.0	n.a.
Real effective exchange rate (REER)	102.9	97.5	102.1	103.4	108.5	n.a.	110.5	112.7	n.a.
2. Financial institutions sector[2/]									
Minimum lending rate (MLR)[3/]	5.9	6.9	7.1	7.0	7.0	n.a.	7.0	7.0	n.a.
12-month fixed deposit rate[3/]	1.0	2.4	2.9	2.5	2.4	n.a.	2.4	2.4	n.a.
Capital adequacy									
Regulatory capital to risk-weighted asset (%)	16.2	15.1	15.9	16.3	15.9	n.a.	16.1	n.a.	n.a.
Earnings and profitability									
Net profit (billion baht)	123.0	146.4	45.2	39.2	52.3	n.a.			
Return on assets (ROA)	1.1	1.2	1.3	1.2	1.5	n.a.			
Liquidity									
Loan to deposit and B/E	88.3	89.9	91.4	93.1	93.4	n.a.	93.5	n.a.	n.a.
3. Household sector									
Debt to financial assets	36.8	41.6	44.9	46.4	46.0	n.a.			
NPL and delinquency ratio (%)									
Financial institutions :									
Personal loans	6.0	5.8	5.8	5.8	6.0	n.a.			
Credit cards	4.2	4.8	4.5	3.9	4.7	n.a.			
Thai commercial banks :									
Consumer loans	4.8	5.2	4.7	4.7	4.9	n.a.			
Mortgage loans	4.1	4.0	3.8	3.7	3.7	n.a.			
Auto leasing	7.3	8.9	7.3	7.4	7.7	n.a.			
4. Non-financial corporate sector [4/]									
Operating profit margin (%)	6.9	6.8	6.8	5.8	6.3	n.a.			
Debt to equity ratio (times)	0.7	0.8	0.9	0.8	0.8	n.a.			
Income coverage ratio (times)	4.7	4.1	3.5	3.7	3.4	n.a.			
Current ratio (times)	1.5	1.5	1.4	1.5	1.5	n.a.			

Table 3.1 Sectoral Indicators for assessing risks and vulnerabilities to financial stability (cont.)

Indicators	2010	2011	2012		2013				
			Q3	Q4	Q1	Q2	Apr	May	Jun
5. Real estate sector									
3-month moving average of the number of approved mortgages from banks (Bangkok and its vicinity)	62,153	62,736	21,636	21,091	19,833	n.a.	5,346	5,030	n.a.
Single-detached and semi-detached houses	18,757	18,991	7,614	6,395	5,538	n.a.	1,622	1,538	n.a.
Townhouses and commercial buildings	21,016	21,813	7,334	7,090	7,421	n.a.	2,016	1,892	n.a.
Condominiums	22,360	21,932	6,688	7,606	6,875	n.a.	1,709	1,600	n.a.
3-month moving average of the number of new openings (Bangkok and its vicinity)	112,269	86,587	21,781	23,865	26,047	n.a.	10,904	10,342	n.a.
Single-detached and semi-detached houses	18,581	20,724	2,954	3,885	2,837	n.a.	1,115	1,171	n.a.
Townhouses and commercial buildings	33,734	25,002	6,941	5,425	4,756	n.a.	2,626	3,254	n.a.
Condominiums	59,954	40,861	11,886	14,555	18,454	n.a.	7,163	5,917	n.a.
Housing price index[5/]									
Single-detached houses (including land)	101.3	104.5	108.7	106.9	108.5	n.a.	107.6	n.a.	n.a.
Townhouses (including land)	103.7	109.7	114.7	113.3	118.4	n.a.	117.4	n.a.	n.a.
Condominiums	115.7	127.1	139.1	136.2	140.9	n.a.	142.2	n.a.	n.a.
Land	113.0	121.7	125.7	129.7	136.3	n.a.	132.1	n.a.	n.a.
6. Fiscal sector									
Public debt to GDP (%)	42.5	40.3	43.9	44.0	44.2	n.a.	n.a.	n.a.	n.a.

[1/] Daily volatility (using exponentially weighted average method)

[2/] Based on data of all commercial banks

[3/] Average value of 4 largest Thai commercial banks

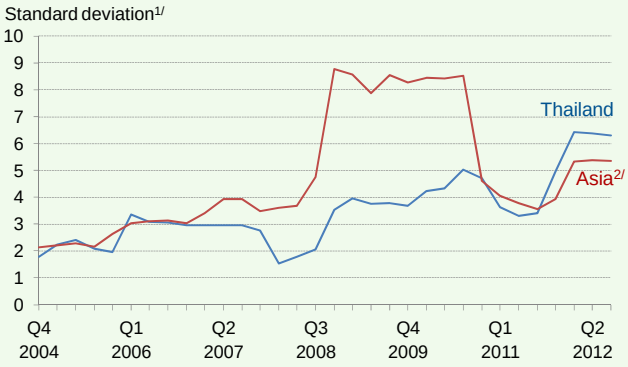
[4/] Only listed companies on SET (median)

[5/] Based on data of new approvals by commercial banks using hedonic regression method (January 2010 = 100)
(Due to the fact that the structure of the housing market has changed significantly, the Bank of Thailand is currently improving the price index to better reflect the structure change)

Why capital flows have become more volatile

In the aftermath of the global crisis, many emerging market economies, including Thailand, have been experiencing higher capital flow volatility **(Chart 1)**. The increase in volatility has been particularly striking during the past three months. At the beginning of the year, emerging market economies were dealing with the large influx of capital inflows led by relatively more promising growth outlook given uncertainties regarding the economic recovery of major economies. In addition, the Bank of Japan's announcement of an aggressive expansion of quantitative easing program flagged another trigger event driving more inflows into emerging markets. Beginning in May, capital flows suddenly reversed their earlier direction upon improvements in US economic data along with an anticipation of a sooner-than-expected QE tapering as well as the recent weakening of Thai economic outlook. In the case of Thailand, reversal of flows together with changing investors' sentiment has caused the Thai baht to so far depreciate by more than 1 percent from the beginning of the year, compared with the peak of 6 percent appreciation at end-April. The Thai baht volatility[1/] has also risen to 7.22 percent in June compared to a mere 2.06 percent at the end of last year.

Chart 1 Volatility of net capital flows



Notes: 1/volatility calculated as 8-quarter rolling standard deviations of net capital flows
2/Asia includes Indonesia, Malaysia, the Philippines, Thailand, and South Korea

Sources: CEIC and IFS, calculations by Bank of Thailand

This trend of increasingly more rapid capital flows in and out of a country could threaten the country's financial and exchange rate stability. It is also interesting to note the mounting synchronicity of capital flows and exchange rates across emerging markets, which implies greater risk of systemic disruption should something go wrong. This box shades light on four potential driving forces behind the more volatile and synchronized capital flows, and how they interact with and reinforce one another, as follows.

(1) More frequent global shocks. This owed to the growing interconnectedness of world economy, more complex technology, the buildup of structural imbalances and fragility especially in major economies along with heightened uncertainties in the aftermath of the global crisis. These factors have contributed to greater sensitivity of investors to new developments in global economy, and thereby resulting in more frequent, widespread global shocks.[2/] A study by Forbes (2012)[3/] confirmed that global factors have become more important determinants of capital flow volatility in the recent period.

(2) The increase in the size of capital flows relative to the size of an economy and their more procyclical behavior. Capital account liberalization in many countries coupled with advancements in communication and technology have resulted in lower international transaction costs, which led to larger capital flows over time. On the other hand, the ease of investment and more rapid spread of news have also led to greater sensitivity of capital flows to both positive and negative economic developments.

On the procyclicality, capital inflows to emerging market economies have generally been found to establish procyclical behavior—plentiful in good times and scarce in bad times. Such behavior can lead to excessive volatiity in the recipient economy. Furthermore, recent data reveals that the procyclicality of capital flows, measured by the difference between the size of foreign flows during high-growth and low-growth periods, has increased in the case of Thailand as well as other Asian economies **(Table 1)**.

1/ Calculated using the exponentially weighted moving average of daily changes of the exchange rate (annualized).

2/ For these reasons, the OECD assessed that the global economy would face more frequent global shocks that can be originated from natural disasters, technological failures, epidemics, or other economic crises. Refer to "Future Global Shocks, Improving Risk Governance", OECD Reviews of Risk Management Policies (2011), for further details.

3/ Forbes, K, and F. Warnock (2012), "Capital flow waves: Surges, stops, flight and retrenchment", *Journal of International Economics* 88, 235-51.

Table 1 Net portfolio inflows to GDP during good times and bad times

Net Portfolio Inflows to GDP (%)				
		Good time (1)	Bad time (2)	Amplitude (1) - (2)
Asia	2000-2007	1.89	0.62	1.27
	2008-2012	3.89	1.03	2.86
Thailand	2000-2007	1.23	-0.49	1.72
	2008-2012	3.34	0.41	2.93

Notes: - Net portfolio inflows to GDP shown in percentage
- Good (Bad) time refers to the period of higher (lower) than average GDP growth
- Asia includes Indonesia, Malaysia, the Philippines, Thailand and South Korea

Sources: CEIC and IFS, calculations by Bank of Thailand

Table 2 Correlation of real GDP growth between Asia, Thailand and major economies

Correlation of %YoY Real GDP Growth[1/]					
		US	Euro area	Japan	China
Asia[2/]	2000-2007	0.64	0.44	0.64	0.41
	2008-2012	0.82	0.77	0.87	0.56
Thailand	2000-2007	0.44	-0.08	0.46	0.21
	2008-2012	0.57	0.50	0.63	0.42

Notes: [1/]calculate correlation of real GDP growth using quarterly data
[2/]Asia includes Indonesia, Malaysia, the Philippines, Thailand and South Korea

Sources: CEIC and calculations by Bank of Thailand

(3) More intense shock transmission as a result of an increase in direct and indirect global connectivity via trade, investment, financial institutions and market expectations. While trade and investment linkages are nothing new, they have recently intensified especially after large economies such as China, Russia, and India opened up more to cross-border trade and investment, along with the increasing exploitation of global supply chains.

A transmission channel that has been highlighted more recently is the investor expectations channel. Through this channel, shock in one country, especially one that is highly connected with others, can affect the economic outlook re-assessment of other countries according to the "Wake-up Calls" hypothesis[4/]. The response of capital flows to shifts in expectations can front run real events, leading to self-fulfilling effects which could materialize more rapidly than that through traditional trade and investment channels. Moreover, a shift in expectations tends to occur simultaneously in countries with similar economic fundamentals, a so-called coordinated panic.[5/]

Together, these factors have resulted in greater business cycle synchronicity, as reflected by higher correlation between regional economies, including Thailand, and major economies since the onset of the global crisis in 2008 **(Table 2)**. The most recent case in point was when the Fed signaled a possibility of a sooner-than-expected QE unwinding which happened to coincide with the slowdown of the Chinese economy. The subsequent market reactions illustrate how the front-running of investor expectations about possible repercussions on other economies could result in such great volatility in financial markets, particularly in terms of sudden outflows from Asian and Latin American economies.

(4) Limited monetary and fiscal policy space and tight credit conditions. The zero-bound interest rates and high public debt, particularly in major economies, act to amplify the effect of the expectations channel. With limitations on policy tools to stabilize growth, investors would tend to anticipate the economic impact of shocks to be more severe. In addition, the tight credit condition in many countries stemmed from problems in the financial sector could further impair the effectiveness of monetary policy transmission.

In sum, the aforementioned factors together have resulted in higher volatility of capital flows in and out of emerging market economies, including Thailand. This highlights the role of global factors as increasingly more important drivers of these flows, especially through the psychological effect during the time of heightened uncertainty. Since these factors are sometimes beyond the control of national authorities, all concerned stakeholders, both public and private, must strive to undertake necessary adjustments to strengthen the resiliency of the economy to cushion against greater volatility brought about by the increasing global interconnectedness.

[4/] Forbes, K. (2012), "Capital flows volatility and contagion: A focus on Asia", manuscript, MIT-Sloan School of Management (November).

[5/] Bacchetta and Wincoop (2013) found that the expectations channel is important in explaining greater cyclical synchronicity of global economies. Refer to "The Great Recession: A Self-Fulfilling Global Panic", NBER (May 2013) for more details.

Commercial banks' liquidity and interest rate adjustment

Large commercial banks have yet to adjust deposit and lending rates following the policy interest rate cut on May 29, 2013. A key reason for commercial banks' decision to postpone rate adjustment is persistently high competition for deposits. Understanding the sources of deposit competition and its effect on commercial banks' liquidity, is therefore crucial in assessing the effectiveness of the monetary policy transmission mechanism.

Assessing commercial banks' liquidity

The definition of "liquidity" varies depending on the specific markets considered. Money market liquidity may be assessed by short-term interbank rates which move closely with the policy interest rate. On the other hand, liquidity in the deposit and lending markets, while influenced by money market liquidity, also depends on demand for deposits and loans, availability of loan supply and commercial banks' business strategies, prevailing degree of competition in the banking industry, as well as other factors. This article focuses on liquidity in deposit and lending markets from commercial banks' perspectives.

A basic measure of liquidity used by many banks is the Loan-to-Deposit Ratio (L/D). An increase in the L/D ratio implies a tightening in liquidity as new loan extension exceeds mobilized deposits. At the end of May 2013, the L/D including Bills of Exchange (B/E) ratio of all commercial banks stood at 93.3 percent, reflecting somewhat tighter liquidity compared to recent years **(Chart 1)**. The L/D ratio is a simple and useful gauge of system-wide liquidity, since loans and deposits constitute a large proportion of financial institutions' balance sheets. At present, loans account for 75 percent of other depository financial institutions' total assets, while deposits account for 65 percent of total liabilities.

Chart 1 Loan-to-Deposit (including Bill of Exchange) ratio of commercial banks*



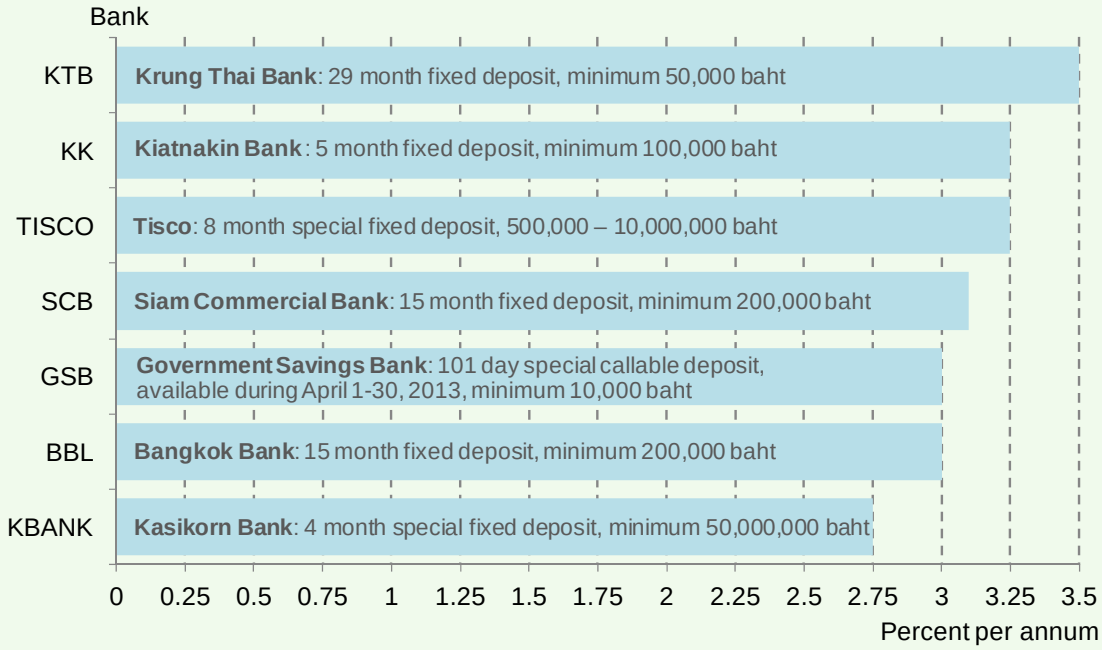
Note: *includes all Thai and foreign commercial banks
Source: Bank of Thailand

The aggregate L/D ratio however needs not represent individual banks' liquidity, which is influenced by a myriad of factors. Deposit competition, business expansion plans, internal liquidity management policy and loan targets for example all have an impact on each institution's liquidity and its outlook. While macro developments such as behavioral shift in savers' demand for deposits or the extent of disintermediation can affect the system's overall liquidity, other micro factors such as deposit competition result in a re-distribution of liquidity within the system–seemingly a zero-sum game. However, shocks to liquidity distribution could still induce financial institutions to intensify competition for deposits, for example by encouraging those short in deposit to engage in active mobilization policy. These behavioral shifts can ultimately affect total deposit and therefore market-wide liquidity.

Financial institutions' competition for deposits

Financial institutions have been fervently competing for deposits in recent years, reflected in part by continued releases of special deposit campaigns paying higher interest rates than those of normal time deposits. Banks' onerous efforts to defend their market shares stem mainly from the following factors : (1) most institutions plan to expand lending at a reasonable pace, and thus are prepared to maintain their deposit base even at higher costs; (2) banks are somewhat concerned that the funding needs related to large-scale public investment projects could trigger another round of deposit competition, especially if SFIs were to be involved; and (3) higher liquidity dispersion among financial institutions along with their different business strategies have played a part in intensifying competition for deposit.

Chart 2 Examples of special deposit products of financial institutions which offer higher interest rates than normal time deposits as of July 4, 2013



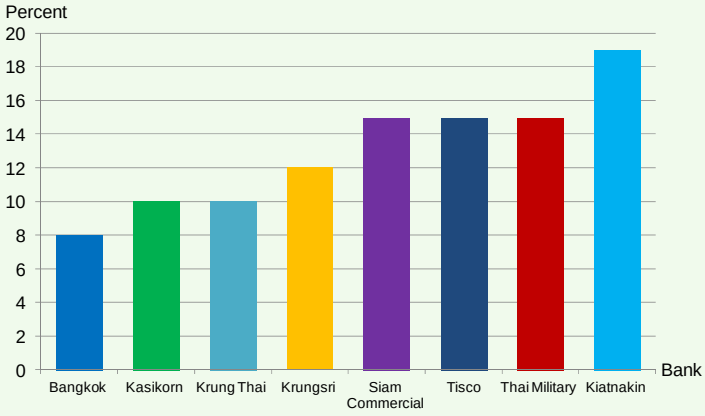
Sources: Commercial banks' websites as of July 4, 2013

Financial institutions that offer special deposit products, especially those with plenty of liquid assets, are incurring short-term costs as these products typically pay higher interest rates than short-term money market rates (close to policy interest rate at 2.50 percent) that these banks earn from their liquid investment **(Chart 2)**. The willingness to bear these short-term costs implies that these banks expect to lend out credits at a reasonably high pace. Over the past two years, the spread between the policy interest rates and the average effective deposit rates has indeed narrowed in line with the high pace of credit growth. A recent survey also indicates that most commercial banks are targeting double-digit loan growth for this year **(Chart 3)**.

SFIs' growth strategies remain another important issue that commercial banks consider in assessing their liquidity outlook. Given the government's plan to undertake large infrastructural projects, including the water management projects and the 2 trillion baht infrastructure project, commercial banks are concerned about potential competition from SFIs to mobillize deposits in order to meet public demand for funds. Commercial banks have stepped up effort to retain their deposit base as a result.

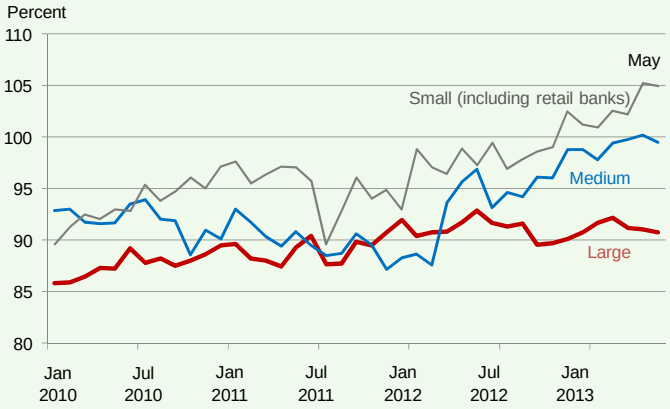
Lastly, business strategies of each individual commercial bank have also affected the competition landscape. During the past couple of years, small and medium-sized commercial banks have been active in extending loans, particularly to auto leasing sector, partly as a result of the government's first-car tax rebate scheme. The resulting higher liquidity dispersion **(Chart 4)** has fuelled greater competition for deposits amongst commercial banks, as swings in liquidity encourages those losing deposits to push and regain market share. Among larger commercial banks themselves, differences in business plans in terms of target growth and strategic loan market segments for example, also promote fierce competition.

Chart 3 Commercial banks' lending targets in 2013



Source: KGI Research June 26, 2013

Chart 4 Loan-to-Deposit (including Bill of Exchange) ratio by size of commercial banks



Source: Bank of Thailand

Summary

The MPC's decision to lower the policy interest rate by 0.25 percent on May 29, 2013 has signaled a more accommodative stance of monetary policy. While the adjustment in monetary policy was fully transmitted to the money market rates, large commercial banks have yet to follow through with reference loan and deposit rates. Anticipating still intense deposit competition, banks are therefore reluctant to adjust deposit rates downward. However, currently tight liquidity in the deposit market does not indicate a lack of liquidity in the money market. Rather, commercial banks' decisions are driven by the desire to defend their deposit base amid active plans to extend loans and dynamic competition for market share. Thus, in this environment, deposit competition is indicative of robust demand for credits as well as banks' willingness to lend, suggesting that financial conditions remain accommodative for the real sector.